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A Project Proposal Report On

KARMASHEEL: DESIGN AND DEVELOPMENT OF A SMART WORKFORCE-MATCHING AND OPPORTUNITY-ADVISORY SYSTEM USING EXPLAINABLE HYBRID RECOMMENDATION TECHNIQUES FOR BLUE-COLLAR AND LOCAL SERVICE WORKERS

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A Project Proposal Submitted To Department Of Electronics And Computer Engineering In Partial Fulfillment Of The Requirement For Bachelor's Degree In Computer Engineering

Department Of Electronics And Computer Engineering

Lalitpur, Nepal

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ABSTRACT

Karmasheel is a proposed web-based workforce-matching and opportunity-advisory platform designed to connect blue-collar workers in Nepal with suitable local employment opportunities and employers. The platform aims to reduce the limitations of informal job-search and hiring methods by providing structured worker and employer profiles, employer verification, job posting, job search, application tracking, automatic CV generation, ratings, complaint handling, and administrative monitoring.

The recommendation module will use a structured skill taxonomy, NLP-assisted skill extraction and normalization for English and Romanized Nepali skill phrases, multi-criteria filtering, cosine similarity as a baseline for comparing standardized skill vectors, Haversine-distance calculation, reciprocal preference scoring, and explainable hybrid weighted ranking. The NLP-assisted normalization process will map different user-entered skill expressions, such as English synonyms and Romanized Nepali terms, to standardized skill tags before matching. The system will recommend relevant jobs to workers and suitable workers to employers while displaying understandable match reasons. An opportunity-advisory feature will analyze near-miss jobs to identify commonly missing skills and suggest practical improvements. The expected outcome is an accessible, transparent, and organized platform that improves the relevance and trustworthiness of blue-collar workforce matching.

Keywords: *Blue-collar workers, cosine similarity, hybrid recommendation, NLP-assisted skill normalization, opportunity advisory, skill taxonomy*

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LIST OF ABBREVIATIONS

API	Application Programming Interface
CV	Curriculum Vitae
DFD	Data Flow Diagram
DRF	Django REST Framework
ER	Entity–Relationship
FP-Growth	Frequent Pattern Growth
NLP	Natural Language Processing
PAN	Permanent Account Number
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SQL	Structured Query Language
UI	User Interface
VAT	Value Added Tax

CHAPTER 1: INTRODUCTION

1.1. Background Information

In Nepal, many blue-collar, local-service, daily-wage, temporary, part-time, and flexible workers still depend on informal methods such as personal contacts, word-of-mouth communication, social-media posts, direct workplace visits, and messaging groups to find employment opportunities. Workers such as electricians, plumbers, cleaners, waiters, delivery workers, event helpers, repair workers, security guards, and students seeking flexible work may not always receive timely information about suitable jobs near their location.

At the same time, employers, households, businesses, and organizations may face difficulty finding skilled, nearby, available, and reliable workers, especially for urgent, short-term, shift-based, or local-service jobs. Existing online job portals are generally more focused on formal and long-term employment. They often depend on detailed resumes, academic qualifications, and lengthy application procedures, whereas blue-collar and local-service work depends more strongly on practical skills, location, availability, work timing, wage expectations, trust, and previous work reliability.

To address this gap, the proposed project introduces Karmasheel, a web-based workforce-matching and opportunity-advisory platform. The system will allow workers to create structured profiles containing their skills, experience, location, availability, preferences, and certification details where applicable. Employers will be able to post job requirements and receive suitable worker recommendations. Similarly, workers will receive relevant job recommendations based on practical matching factors.

The proposed system will use a structured skill taxonomy, NLP-assisted skill normalization, multi-criteria filtering, cosine-similarity-based skill comparison, Haversine-distance calculation, reciprocal preference scoring, and explainable hybrid weighted ranking. The platform will also provide understandable match reasons, automatic CV generation, employer verification, application tracking, ratings, complaints, admin monitoring, and opportunity-advisory suggestions. The aim of Karmasheel is to make blue-collar and local-service workforce matching more organized, transparent, accessible,

and practical.

1.2. Problem Statement

Blue-collar and local-service job matching in Nepal is still largely dependent on informal and unstructured methods. This creates problems for both workers and employers. The major problems addressed by the proposed system are listed below.

Problems Faced by Workers

- Workers often depend on personal contacts, word-of-mouth communication, social-media posts, direct workplace visits, and messaging groups to find jobs.
- Suitable nearby jobs may remain hidden from workers because local, temporary, part-time, urgent, and shift-based opportunities are not organized in a single accessible platform.
- Many workers do not have structured digital profiles to clearly present their skills, experience, availability, preferred work type, expected wage range, and travel preference.
- Many blue-collar and local-service workers may not have a formal CV, which can make it harder for them to present their work experience professionally.
- Workers may find it difficult to identify trustworthy employers because employer information, verification status, previous ratings, and job details may not be clearly available.
- Workers may not know which missing skills are preventing them from qualifying for better or nearby job opportunities.

Problems Faced by Employers

- Employers, households, businesses, and organizations may struggle to find suitable workers quickly, especially for urgent, short-term, shift-based, or local-service jobs.
- Employers may not have a reliable way to compare workers based on practical skills, experience, location, availability, wage expectations, and reliability.

- Employers may receive applications from unsuitable candidates because job requirements and worker profiles are not always matched in a structured way.
- Employers may not know whether a worker is available on the required date and time or whether the job location is practical for the worker.
- Employers may have limited information about worker reliability, previous work performance, ratings, and complaint history.

1.3. Objectives

The objectives of this project are:

- To develop a web-based platform that supports worker and employer profiles, job posting, job searching, applications, employer verification, CV generation, ratings, complaints, and admin monitoring.
- To provide explainable worker-to-job and job-to-worker recommendations using NLP-assisted skill normalization, standardized skills, filtering, skill matching, location scoring, and opportunity-advisory suggestion

1.4. Scope and Applications

The project focuses on developing the first functional version of the Karmasheel platform for workforce matching and opportunity advisory. The scope is divided into features included in the first version and features excluded from the first version.

Included in the Scope

- Develop worker and employer registration, login, profile management, and employer-verification features.
- Support job posting, job searching, filtering, application submission, and application-status tracking.
- Provide worker-to-job and job-to-worker recommendations using skills, location, availability, experience, preferences, and reliability indicators.
- Include NLP-assisted skill extraction and normalization for common English and Romanized Nepali skill phrases during worker-profile creation and job posting.

- Use a structured skill taxonomy to map different skill expressions to standardized skill tags.
- Provide explainable match reasons so that workers and employers can understand why a job or worker is recommended.
- Generate automatic CVs from worker-profile data and support ratings, reviews, complaints, and rehire-related features.
- Include an opportunity-advisory module that identifies near-miss jobs, missing skills, and practical skill-improvement suggestions.
- Provide an admin dashboard for monitoring profiles, jobs, verification requests, complaints, skill categories, and unmatched skill terms.

Excluded from the Scope

- The first version will not guarantee employment or hiring decisions; it will only support matching, recommendation, and application management.
- The system will not include online payment, wage transfer, or financial transaction handling.
- The system will not perform fully automated government-level employer verification. Employer verification will initially depend on rule-based format checking and admin review.
- The first version will not use a complex deep-learning recommendation model or large-scale trained NLP model. Advanced semantic embeddings may be added later after collecting sufficient real-world data.
- The NLP component will not provide full translation or perfect understanding of all Nepali language variations. Low-confidence or unmatched skill terms will be forwarded for admin review.
- The first version will not include a dedicated mobile application. The system will be developed as a responsive web-based platform.

1.5. Applications

The proposed Karmasheel platform can be applied in different local employment and service-matching situations. Its main applications are as follows:

- **Local workforce matching:** The platform can help connect blue-collar and local service workers with nearby employers based on practical requirements such as skills, location, availability, experience, and wage preferences.
- **Short-term and flexible job discovery:** Workers looking for daily-wage, temporary, part-time, event-based, or shift-based jobs can use the system to find relevant opportunities more easily.
- **Employer-side worker recommendation:** Employers, households, businesses, and organizations can use the platform to find suitable workers for urgent, temporary, or regular work requirements.
- **Skill-gap identification:** The opportunity-advisory module can help workers understand which missing skills may improve their chances of qualifying for more jobs.
- **Structured worker-profile and CV support:** Workers who do not have formal resumes can use the platform to maintain structured profiles and automatically generate simple CVs.
- **Trust and verification support:** Employer verification, ratings, reviews, complaint handling, and admin monitoring can help improve trust between workers and employers.
- **Labour-market analysis:** The admin dashboard can provide useful insights such as active job categories, frequently missing skills, jobs with no suitable workers, and common unmatched skill terms.
- **Future expansion for training and employment support:** In later versions, the platform can support vocational training recommendations, partnership with training centers, and improved semantic matching using real-world data.

CHAPTER 2: LITERATURE REVIEW

The development of workforce-matching platforms has become an important research area because job seekers and employers often struggle to find suitable matches efficiently. Existing studies have explored content-based filtering, hybrid recommender systems, semantic similarity, natural-language processing, occupational-skill normalization, knowledge graphs, explainable recommendations, and reciprocal matching. These approaches provide useful ideas for developing Karmasheel, a workforce-matching and opportunity-advisory platform for blue-collar workers.

[1] Ertugrul and Bitirim presented a systematic literature review of job recommender systems. The study examines major recommendation approaches, including content-based filtering, collaborative filtering, knowledge-based recommendation, context-aware techniques, and hybrid systems. It also discusses applications, evaluation methods, open issues, and research challenges. The review is important for Karmasheel because it shows that job matching should not rely on a single factor. However, many existing systems assume access to extensive user histories, detailed resumes, and large recruitment datasets, which may not be available during the initial stage of a new platform.

[2] Cano and Morisio reviewed hybrid recommender systems and explained how multiple recommendation techniques can be combined to reduce the limitations of individual methods. Hybrid recommendation is particularly useful when a system faces cold-start conditions or data sparsity. This is relevant to Karmasheel because the platform will initially have limited historical data about applications, hiring outcomes, ratings, and user interactions. Therefore, the proposed system will begin with a transparent hybrid method based on structured skills, distance, availability, experience, preferences, and reliability indicators. A limitation is that selecting suitable weights for different factors requires testing and adjustment.

[3] Manning, Raghavan, and Schutze described vector-space scoring and cosine similarity in information retrieval. Cosine similarity measures the similarity between two vectors according to the angle between them. For Karmasheel, each vector dimension can represent a standardized skill. A worker-skill vector can then be compared with a job-skill vector to calculate a baseline similarity score. This method is simple, inter-

pretable, and does not require model training. However, cosine similarity alone does not account for important contextual factors such as worker availability, geographical distance, wage expectations, certification, and reliability.

[4] Zhao *et al.* proposed an embedding-based recommender system for large-scale job-to-candidate matching at CareerBuilder. Their system follows a two-stage architecture. In the first stage, raw text, semantic entities, and location information are combined into fused embeddings. A Faiss index is then used to retrieve a smaller set of potentially relevant candidates through approximate nearest-neighbor search. In the second stage, the retrieved candidates are reranked using additional contextual information. The study demonstrates how large recruitment platforms can use text, skill, and location information together for efficient matching at scale. This supports the idea that free-text job and worker information can contain useful semantic signals. However, the complete architecture requires large datasets, deep-learning models, and specialized indexing infrastructure. For the initial version of Karmasheel, the same two-stage idea will be simplified into NLP-assisted skill normalization, multi-criteria filtering, and explainable weighted ranking instead of deep-learning embeddings and Faiss-based retrieval.

[5] Javed *et al.* presented SKILL, a Named Entity Normalization system for occupational skills. The system uses a skill tagger to identify and normalize skill phrases from resumes and job advertisements. It also includes a skill-sense disambiguation component to determine the correct meaning of an identified phrase. This work is important because workers and employers may use different expressions for similar practical skills. For example, 'home wiring," 'residential wiring," and 'house electrical installation" may refer to the standardized skill tag 'house wiring." In Karmasheel, this idea is extended to common English and Romanized Nepali skill phrases, such as 'ghar wiring," 'fan jodne," and "pipe leak banaune." The complete SKILL system uses advanced semantic techniques and requires substantial training data. Karmasheel will not implement a large-scale trained skill-normalization model in the first version. Instead, it will use lightweight NLP-assisted processing, including lowercasing, punctuation removal, tokenization, synonym lookup, Romanized Nepali alias mapping, fuzzy matching, and admin-reviewed taxonomy updates.

[6] De Groot, Schutte, and Graus proposed a job-posting-enriched Skills and Occupation Knowledge Graph. Their approach combines existing skill and occupation taxonomies

with information collected from job advertisements. The study explores link prediction for measuring the relevance of skills to occupations, node-similarity methods, shortest-path algorithms for career pathfinding, and term weighting for identifying distinctive skills. This work is relevant to the opportunity-advisory feature of Karmasheel because a worker may benefit from understanding which missing skills could improve access to additional opportunities. However, constructing and maintaining a detailed knowledge graph requires a sufficiently large and reliable dataset. The first prototype will instead use a structured taxonomy and near-miss job analysis.

[7] Ong and Lim developed SkillRec, a data-driven system for job-skill recommendation and career insights. Their method processes job descriptions, represents job titles using word or sentence embeddings, and uses a feed-forward neural network to recommend relevant skills. The approach shows how job-posting data can support practical skill suggestions. This is closely related to the opportunity advisor in Karmasheel. However, the initial prototype will not train a neural network. It will first identify frequently missing skills from relevant near-miss jobs and rank them using skill importance, category relevance, and location relevance.

[8] Zhang and Chen reviewed explainable recommender systems and emphasized that users should receive understandable reasons in addition to ranked recommendations. Explainability can improve transparency, trust, user satisfaction, and system debugging. This principle is important for Karmasheel because workers and employers should understand why a match is recommended. For example, an employer may see that a worker matches four out of five required skills, is located within the preferred distance, and is available on the required date. A limitation is that highly complex models may require additional explanation techniques. Karmasheel will begin with a transparent weighted formula and template-based reasons.

[9] Palomares *et al.* analyzed reciprocal recommender systems, in which both sides of a recommendation must find the proposed match acceptable. Recruitment differs from product recommendation because a worker is not merely an item to be ranked. The worker also has preferences and constraints. This concept is relevant to Karmasheel because an employer may prefer a worker with suitable skills and experience, while the worker may prefer a job with an acceptable wage, schedule, work type, and travel distance. The limitation is that balancing both perspectives requires careful weight

selection and user testing.

[10] The Nepal Labour Force Survey 2017/18 reported that the informal sector accounted for 62.2% of employment and that 84.6% of employed people were in informal employment. These figures represent different concepts: informal-sector employment refers to the type of establishment, whereas informal employment is a broader measure related to employment conditions. The findings show the relevance of developing a more organized platform for workers who may currently depend on informal channels such as personal contacts, word-of-mouth communication, direct workplace visits, and social-media posts.

Despite the contributions of the reviewed studies, several limitations remain. Many existing approaches are designed for large recruitment platforms and depend on extensive datasets, detailed resumes, historical interactions, semantic models, or advanced infrastructure. These assumptions may not be suitable for an initial prototype targeting local and blue-collar employment in Nepal.

Karmasheel addresses this gap by using an achievable and transparent approach. The first version will combine standardized skill tags, NLP-assisted skill extraction and normalization, English and Romanized Nepali alias mapping, synonym mapping, multi-criteria filtering, cosine similarity as a baseline, Haversine-distance calculation, availability checking, reciprocal preferences, and explainable weighted ranking. It will also analyze near-miss jobs to suggest practical skill improvements. Advanced semantic embeddings, advanced automatic skill extraction, knowledge-graph learning, Faiss-based retrieval, and interaction-based personalization will remain future enhancements after sufficient real-world data has been collected.

CHAPTER 3: REQUIREMENT ANALYSIS AND SYSTEM DESIGN

3.1. Requirement Analysis

3.1.1. Functional Requirements

Figure 3.1 shows the use-case diagram of the proposed system.

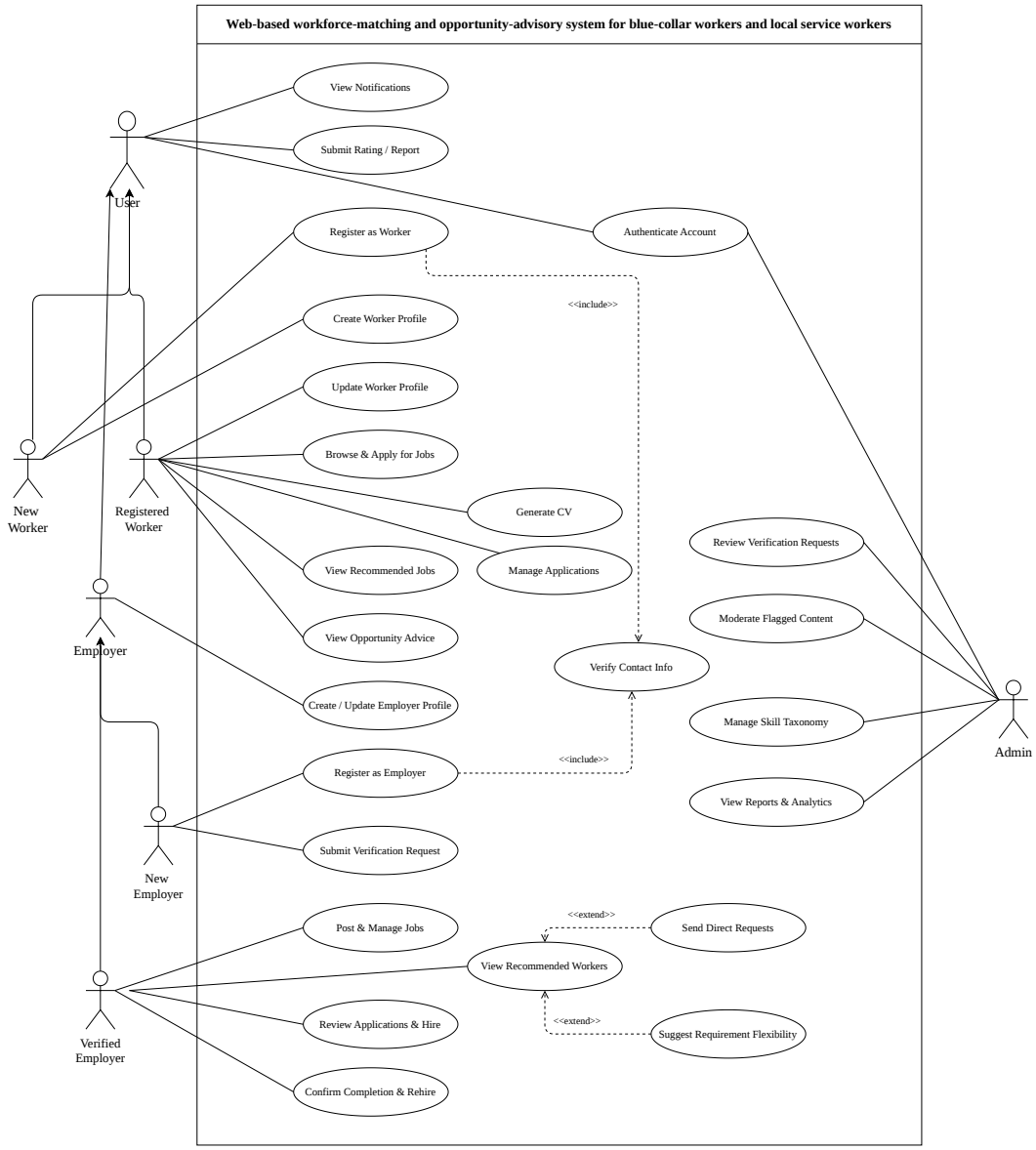


Figure 3.1: Use-case diagram of the Karmasheel platform

From figure 3.1, the general user actor represents common platform actions. A user can log in or log out, receive notifications, submit ratings after job completion, and report suspicious activities. A new worker can register, verify contact information, and create a worker profile. An registered worker can update the profile, browse and apply for jobs, view recommended jobs, manage applications, generate a CV, and view opportunity advice.

A new employer can register and verify contact information. An employer can create or update a profile and submit a verification request. After approval, a verified employer can post jobs, view recommended workers, review applications, hire workers, send direct requests, confirm job completion, and save reliable workers for rehiring. The admin can review verification requests, moderate flagged content, manage the skill taxonomy, and view reports and analytics.

Table 3.1 summarizes the functional requirements.

Table 3.1: Functional Requirements

ID	**Requirement**	**Description**
FR1	User registration	The system shall allow new workers and new employers to create accounts using the required registration information.
FR2	Contact verification	The system shall verify the contact information provided during worker and employer registration.
FR3	Authentication	The system shall allow users and the admin to log in and log out securely.
FR4	Worker-profile management	The system shall allow workers to create and update profiles containing selected skills, entered skill phrases, experience, location, availability, travel radius, wage expectations, and optional certification details.
FR5	Employer-profile management	The system shall allow employers to create and update profiles containing employer details, location, contact information, and verification information.

ID	Requirement	Description
FR6	Employer-verification request	The system shall allow employers to submit verification requests for admin review.
FR7	Job posting and management	The system shall allow verified employers to create, update, and close job posts containing job title, required skills, preferred skills, entered skill phrases, location, schedule, duration, wage range, and other relevant information.
FR8	Job browsing and application	The system shall allow registered workers to browse available jobs, apply filters, and submit applications.
FR9	Recommended jobs	The system shall display ranked job recommendations to workers based on skills, location, availability, experience, preferences, and reliability indicators.
FR10	Application management	The system shall allow workers to track application statuses such as applied, shortlisted, contacted, hired, rejected, withdrawn, and completed.
FR11	Automatic CV generation	The system shall generate a downloadable CV using information stored in the worker profile.
FR12	Opportunity advisory	The system shall identify near-miss jobs and suggest practical skills that may help workers qualify for additional opportunities.
FR13	Recommended workers	The system shall provide verified employers with ranked worker recommendations for their job posts.
FR14	Direct worker requests	The system shall allow verified employers to send direct requests to suitable workers after reviewing recommended profiles.
FR15	Application review and hiring	The system shall allow verified employers to review applications, shortlist workers, contact applicants, and record hiring decisions.

ID	Requirement	Description
FR16	Completion and rehiring	The system shall allow verified employers to confirm job completion and save reliable workers for future rehiring.
FR17	Notifications	The system shall notify users about important events such as applications, shortlisting, direct requests, hiring decisions, and verification updates.
FR18	Ratings	The system shall allow workers and employers to submit ratings after a job has been marked as completed.
FR19	Complaints and reports	The system shall allow users to report suspicious jobs, incorrect information, payment-related issues, or inappropriate behavior.
FR20	Verification-request review	The system shall allow the admin to approve, reject, or request clarification for employer-verification requests.
FR21	Flagged-content moderation	The system shall allow the admin to inspect reported job posts, complaints, and suspicious user activities.
FR22	Skill-taxonomy management	The system shall allow the admin to manage categories, subcategories, standardized skill tags, English synonyms, Romanized Nepali aliases, and unmatched skill terms.
FR23	Reports and analytics	The system shall allow the admin to view usage metrics such as active workers, jobs posted, application rates, hire rates, commonly missing skills, and frequently unmatched skill terms.
FR24	Requirement-flexibility suggestion	The system shall suggest optional requirement adjustments when a job post has few or no eligible workers. The employer shall remain responsible for deciding whether to apply the suggestion.

ID	Requirement	Description
FR25	NLP-assisted skill normalization	The system shall process English and Romanized Nepali skill phrases entered by workers and employers and map them to standardized skill tags using text preprocessing, synonym lookup, alias mapping, fuzzy matching, and admin-reviewed taxonomy records.

3.1.2. Non-Functional Requirements

The non-functional requirements specify the expected quality of the platform. Table 3.2 summarizes these requirements.

Table 3.2: Non-Functional Requirements

ID	Requirement	Description
NFR1	Usability	The system should provide clear navigation, understandable labels, structured forms, skill-selection options, and simple input fields suitable for users with limited experience using digital platforms.
NFR2	Performance	The system should retrieve filtered results, normalize skill phrases, and generate ranked recommendations within a reasonable response time. A provisional target is five seconds or less under the expected pilot workload.
NFR3	Security	The system should securely store passwords using hashing and enforce authentication and role-based authorization for worker, employer, and admin functions.
NFR4	Privacy	The system should collect only the information required for matching, verification, and communication. Sensitive information should not be exposed publicly without permission.

ID	Requirement	Description
NFR5	Reliability	The system should preserve consistent records for profiles, applications, ratings, verification requests, job-completion statuses, and skill-normalization records. Invalid status transitions should be prevented.
NFR6	Maintainability	The backend should follow a modular structure so that profile management, recommendation logic, taxonomy management, NLP-assisted skill normalization, and verification workflows can be updated independently.
NFR7	Scalability	The system should support the initial pilot dataset while allowing additional workers, employers, job categories, skill aliases, and future recommendation improvements to be added gradually.
NFR8	Explainability	The system should display understandable reasons for recommendations, such as matched skills, distance, availability, experience, preferences, and verification status.
NFR9	Compatibility	The web-based platform should work on commonly used modern browsers and adapt to desktop and mobile-screen sizes.
NFR10	Language robustness	The system should handle common English and Romanized Nepali variations of skill terms and reduce duplicate skill records by mapping similar expressions to standardized skill tags. Low-confidence or unmatched terms should be available for admin review.

3.1.3. Technical Feasibility

The proposed system is technically feasible within the available development period because the initial version does not require a complex machine-learning model or large-

scale artificial-intelligence infrastructure. The platform can be developed using Python, Django, Django REST Framework, PostgreSQL, HTML, CSS, and JavaScript.

The recommendation module will begin with structured data and transparent algorithms. Candidate filtering will use category, availability, certification, and location conditions. Before skill matching, the system will apply NLP-assisted skill extraction and normalization to process common English and Romanized Nepali skill phrases entered by workers and employers. This normalization process will use lightweight and explainable techniques such as lowercasing, punctuation removal, tokenization, synonym lookup, alias mapping, fuzzy string matching, and taxonomy-based mapping.

After normalization, user-entered skill phrases will be mapped to standardized skill tags. Cosine similarity will then be used as a baseline for comparing standardized worker-skill and job-skill vectors. The final ranking will also consider distance, experience, preferences, availability, and reliability indicators through a weighted scoring method. The Haversine formula will be used to calculate geographical distance between workers and job locations.

The NLP-assisted normalization module will not require training a large language model in the first version. Low-confidence or unmatched skill terms will be stored for admin review and later added to the skill taxonomy or alias list when appropriate. This makes the approach achievable for the proposal stage while allowing gradual improvement as more real user data is collected.

Employer verification will initially use rule-based format validation followed by admin review. Automatic CV generation will use a predefined template filled with worker-profile data. These methods are achievable within the project timeline using the selected software tools and an incremental development approach.

3.1.4. Operational Feasibility

The platform addresses practical problems faced by both workers and employers. Workers can create structured profiles, discover nearby opportunities, generate CVs, and receive skill-gap suggestions. Employers can publish job requirements, identify available workers, and review ranked recommendations.

The use of structured forms and standardized skills reduces the difficulty of writing detailed descriptions. Admin moderation, employer-verification requests, complaint

handling, and flagged-content review improve trust and operational control.

3.1.5. Economic Feasibility

The first version does not require specialized hardware. The proposed software stack is based mainly on open-source technologies. Development and testing can be performed using existing computers, GitHub, Postman, and a local server or an affordable cloud platform for demonstration.

3.1.6. Schedule Feasibility

The project is feasible within the available development period if the team implements the system incrementally. Essential features such as authentication, profiles, job posting, applications, and filtering should be developed first. Recommendation ranking, CV generation, employer verification, ratings, opportunity advisory, and analytics can then be added gradually.

3.1.7. Legal and Ethical Feasibility

The project will use data responsibly for development, testing, and demonstration. Initial datasets may include suitably licensed public datasets, consent-based data collection, and clearly labelled synthetic records. Data from third-party websites will be used only with permission and according to their terms of use. Personal information collected from users will be limited to necessary fields and protected from unauthorized access.

3.2. System Design

3.2.1. Data Flow Diagram (DFD Level 0 and Level 1)

For the Karmasheel platform, two levels of DFD are presented. The Level-0 DFD shows the entire platform as a single process. The Level-1 DFD expands that single process into three major sub-processes and shows the principal data stores used by the system.

DFD Level 0

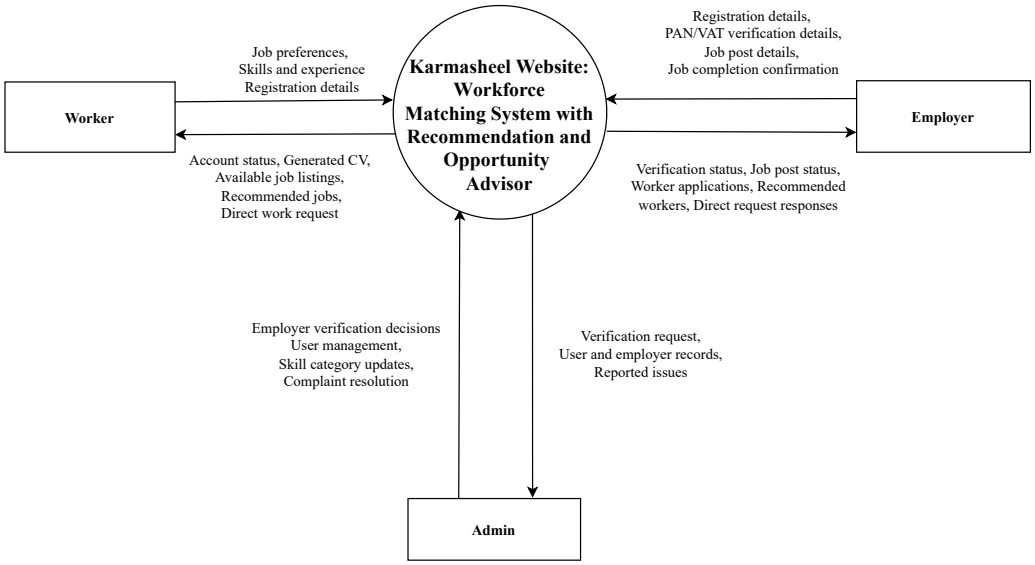


Figure 3.2: DFD Level 0 of the Karmasheel platform

The Level-0 DFD presents the Karmasheel website as a single high-level process: Workforce-Matching System with Recommendation and Opportunity Advisor. The diagram shows the interaction of the system with three external entities: the worker, the employer, and the admin.

A worker provides registration details, skills, experience, and job preferences. In response, the platform provides the worker with account-status updates, generated CVs, available job listings, recommended jobs, and direct work requests.

An employer provides registration details, PAN or VAT verification details, job-post information, and job-completion confirmations. The system returns verification-status updates, job-post status information, worker applications, recommended workers, and responses to direct requests.

The admin receives verification requests, user and employer records, and reported issues from the system. After reviewing the submitted information, the admin provides employer-verification decisions, user-management actions, skill-category updates, and complaint-resolution decisions.

As shown in Figure 3.2, the Level-0 DFD provides an abstract view of the platform boundary. It does not describe internal modules or database tables. Its purpose is to

show the major information exchanged between the Karmasheel system and its external users.

DFD Level 1

The Level-1 DFD decomposes the single Karmasheel process shown in the Level-0 DFD into three major processes:

1. User Profile and Verification Management.
2. Job Matching and Recommendation Management.
3. Application and Work Management.

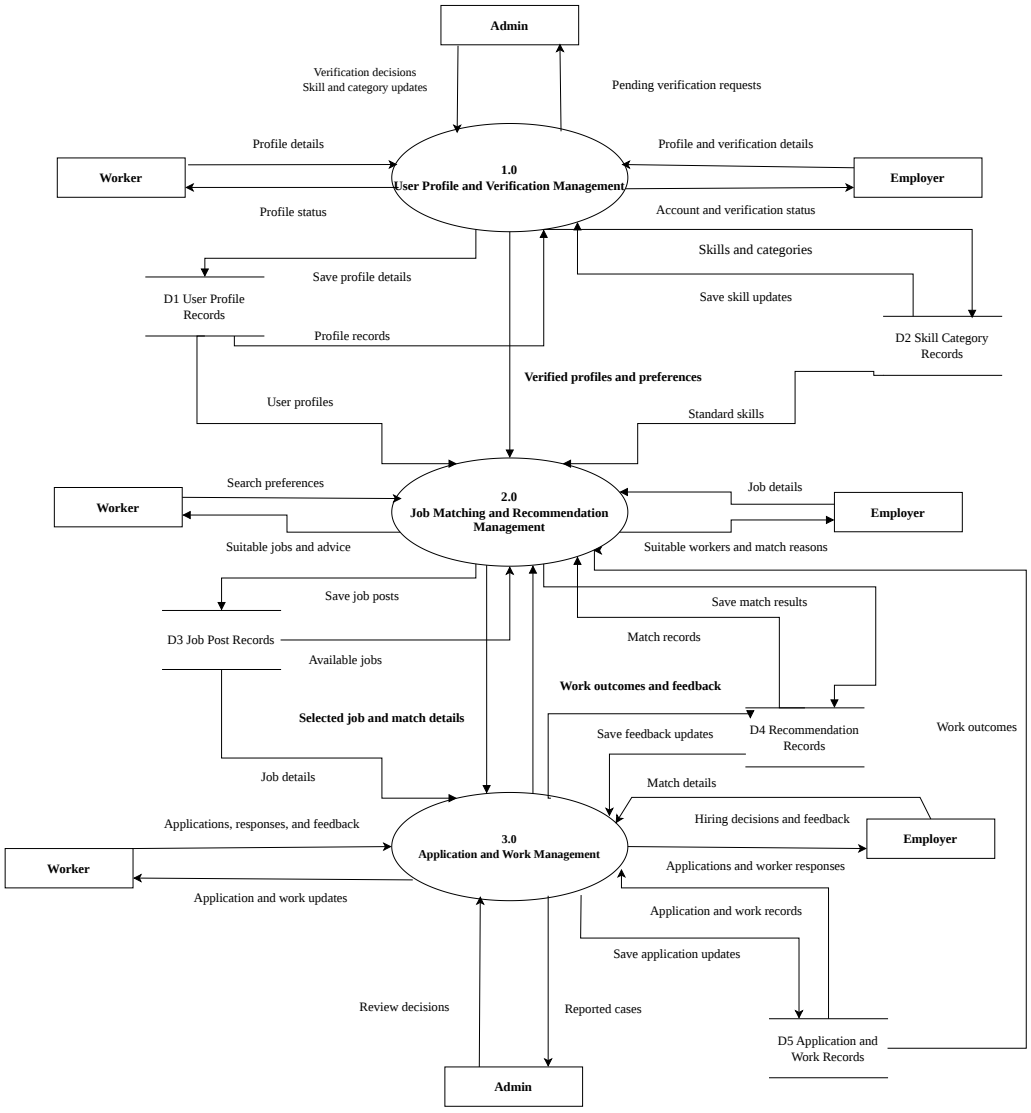


Figure 3.3: DFD Level 1 of the Karmasheel platform

Figure 3.3 shows how the three internal processes exchange information with users and data stores. The following sections explain the responsibility of each process.

Process 1.0: User Profile and Verification Management The first process manages worker profiles, employer profiles, verification requests, and skill-category information. Workers submit profile details, while employers submit profile and verification details. The process returns profile status information to workers and account or verification-status information to employers.

The admin receives pending verification requests and provides verification decisions or skill-category updates. Profile information is saved in D1 User Profile Records. Skill

and category information is stored in D2 Skill Category Records. Verified profiles and user preferences are then passed to the job-matching process.

Process 2.0: Job Matching and Recommendation Management The second process manages job searching, job posting, matching, and recommendation generation. Workers submit search preferences and receive suitable jobs and opportunity advice. Employers submit job details and receive suitable-worker recommendations with understandable match reasons.

The process retrieves worker profiles from D1 User Profile Records, standardized skills from D2 Skill Category Records, and available jobs from D3 Job Post Records. Newly created job posts are saved in D3 Job Post Records. Generated match results are stored in D4 Recommendation Records.

After matching has been completed, selected job and match details are forwarded to the application-and-work-management process.

Process 3.0: Application and Work Management The third process manages applications, employer responses, hiring decisions, job-completion records, ratings, feedback, and reported cases. Workers submit applications, responses, and feedback. The process returns application-status and work-status updates to workers.

Employers receive applications and worker responses. They submit hiring decisions and feedback after reviewing applicants or completing a job. Application and work records are saved in D5 Application and Work Records. The process also retrieves match details from D4 Recommendation Records and updates feedback-related information when work has been completed.

Reported cases are forwarded to the admin for review. After examining the issue, the admin provides a review decision. Work outcomes and feedback are also passed back to the recommendation process so that recommendation records can be updated.

The Level-1 DFD maintains a clear separation of responsibilities. Profile and verification information is handled by Process 1.0, matching and recommendations are handled by Process 2.0, and applications and completed-work records are handled by Process 3.0.

The diagram also avoids dead-end data flows. Each data store receives information from a process and returns relevant records when required. This reflects the practical

behavior of the system: records are not only stored but are also retrieved and used by the appropriate modules.

3.2.2. Workflow Diagrams

The following user-workflow diagrams provide a role-based view of the operations performed through the Karmasheel platform. Unlike the Data Flow Diagrams, which focus on the movement of information between processes, external entities, and data stores, these diagrams describe the sequence of actions visible to the users of the platform.

Two workflow diagrams are included. The first diagram presents the administrative workflow. The second diagram combines the worker and employer workflows to show how both actors participate in the job-matching process from registration to job completion and feedback.

Admin Workflow

The admin workflow describes the major activities performed through the admin dashboard. The admin begins by logging into the system and accessing the dashboard. From the dashboard, the admin selects an administrative task. The available tasks include reviewing employer-verification requests, reviewing flagged cases, viewing reports and analytics, and managing categories and skills.

When reviewing an employer-verification request, the admin examines the submitted employer details and decides whether the request should be approved, rejected, or returned for clarification. The system then updates the verification status accordingly.

When reviewing flagged cases, the admin examines the case details and the reason for the flag. A flagged case may involve repeated complaints, suspicious accounts, misleading job information, or reported job posts. If no action is required, the admin marks the case as reviewed. Otherwise, the admin selects an appropriate action, such as issuing a warning, restricting an account, suspending an account, removing inappropriate content, or correcting information. The decision is then recorded by the system.

The admin can also view reports and analytics to monitor the overall condition of the platform. The displayed information may include the number of active workers, verified employers, job posts, applications, hires, complaints, and frequently missing skills.

The category-and-skill management workflow allows the admin to view existing entries and add, edit, or remove approved categories, subcategories, skill tags, and alternative skill terms. This is important because the structured skill taxonomy is used during worker-profile creation, job posting, matching, and opportunity advisory.

After completing an administrative task, the admin can return to the dashboard and select another task or log out of the system.

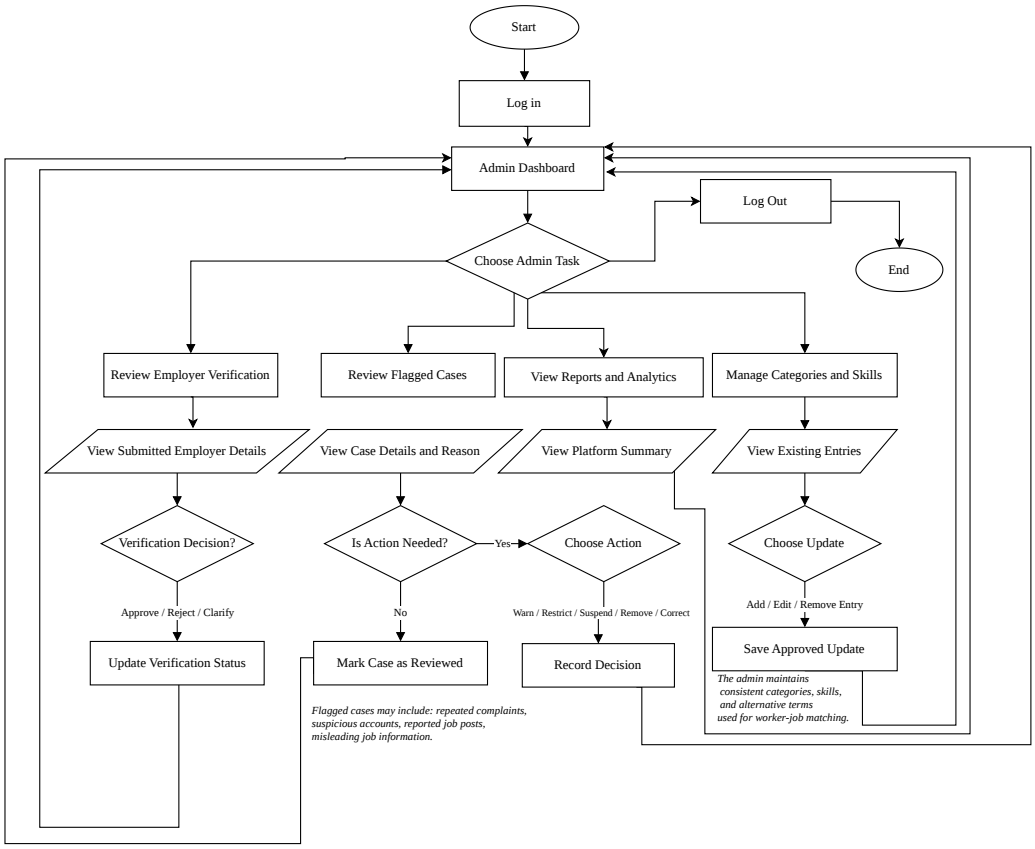


Figure 3.4: Admin workflow of the Karmasheel platform

As shown in Figure 3.4, the admin dashboard acts as the central point for administrative operations. Each completed activity returns the admin to the dashboard so that another task can be selected. This prevents dead-end paths and represents the repeated nature of platform monitoring and maintenance.

Worker and Employer Workflow

The worker-and-employer workflow presents the connected activities of the two main platform users. It begins when a user selects a user type: worker or employer. The

workflow then separates into two branches and later reconnects when a worker is hired and the work is performed.

On the worker side, a new worker registers and creates a structured profile. The profile may include skills, experience, location, availability, preferred work type, and expected wage range. A registered worker can log in and explore available opportunities. The worker may search for jobs, apply filters, view recommended jobs, and inspect job details.

When the worker is interested in a suitable job, the worker submits an application and tracks its progress. The application status may change through stages such as applied, shortlisted, accepted, rejected, or completed. The worker can also generate a CV automatically from the stored profile data and view skill-improvement suggestions generated by the opportunity-advisory module.

A worker may also receive a direct job request from an employer. The worker can accept or reject the request. If the request is accepted, the worker proceeds with the job. If the request is rejected, the worker can return to the job exploration process.

On the employer side, a new employer registers, creates a profile, and submits verification details. The employer must wait for the admin to review the submitted information. If the details are rejected or require clarification, the employer updates or resubmits the information. Once verified, the employer can post a job requirement.

A job requirement may include the required and preferred skills, job location, work schedule, wage range, and number of workers needed. After the job is posted, the employer views suitable workers recommended by the platform. The employer may also review applicants and inspect why each worker is considered suitable.

If no suitable worker is found, the platform may suggest optional changes to the job requirements. For example, it may recommend moving a less essential skill from the required list to the preferred list. The employer remains responsible for deciding whether the suggestion should be accepted.

When a suitable worker is found, the employer shortlists or selects the worker. The employer may hire a worker by accepting an application or by sending a direct request. After the worker is hired, the work is performed.

When the work has been completed, the employer and worker confirm completion and

rate each other. The employer may also save a reliable worker to a trusted list for future jobs. The workflow then ends.

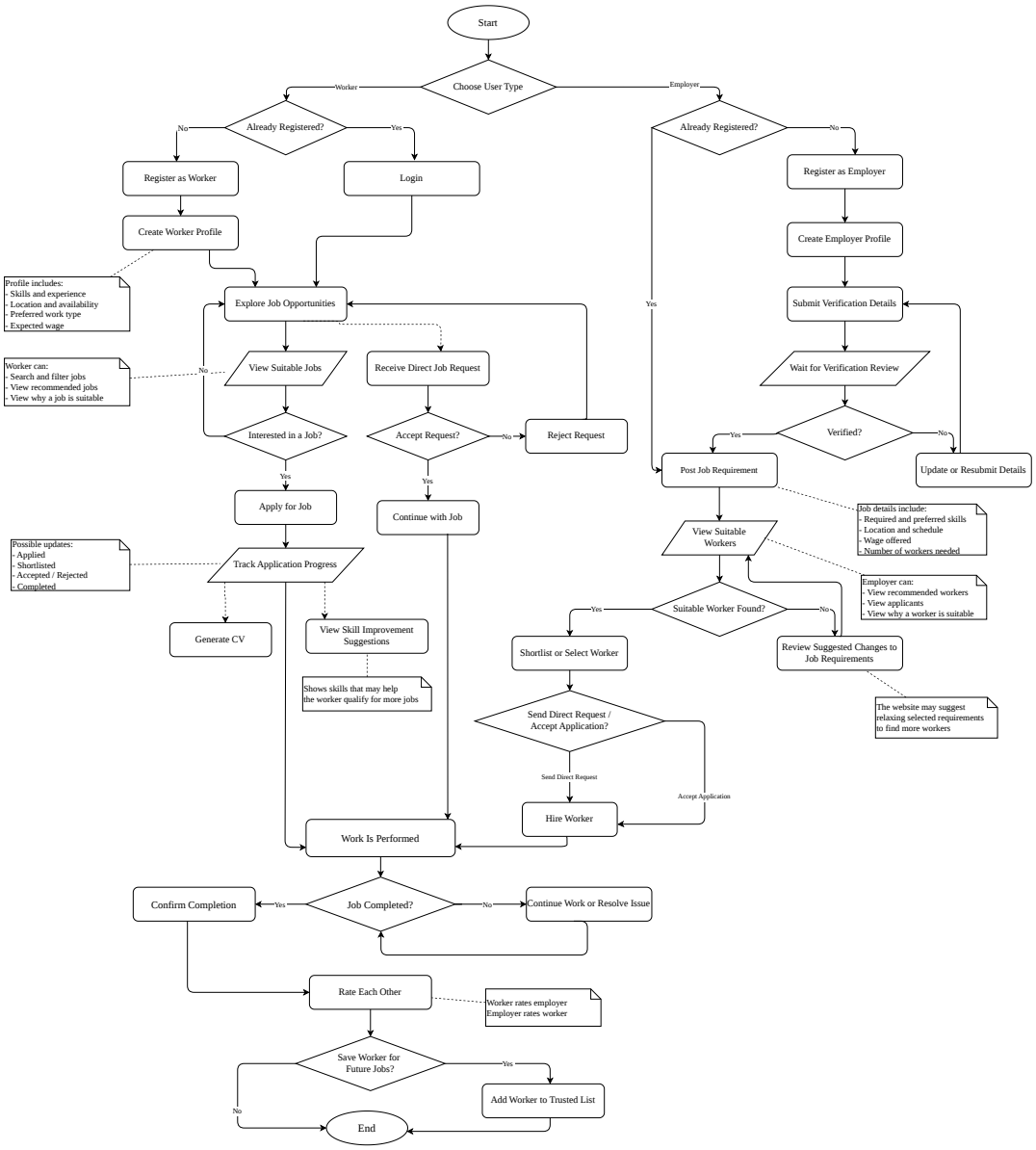


Figure 3.5: Worker and employer workflow of the Karmasheel platform

As illustrated in Figure 3.5, the worker and employer activities are connected through the job-matching process. The platform does not merely provide a job-posting facility. It supports profile creation, employer verification, job exploration, recommendations, applications, direct requests, hiring, work completion, ratings, and trusted worker rehiring. The diagram also shows that workers and employers can return to earlier activities when a job, request, or candidate is not suitable.

CHAPTER 4: METHODOLOGY

This chapter describes the proposed implementation methodology for the Karmasheel platform. It explains the system architecture, structured data collection, skill taxonomy, employer-verification workflow, multi-criteria filtering, hybrid recommendation technique, application tracking, automatic CV generation, ratings, complaint handling, incremental development model, and tools used during implementation.

4.1. Basic System Architecture

The platform is designed using a layered architecture to organise its main components clearly and efficiently. The architecture separates user interfaces, application modules, shared services, and data storage. This structure helps simplify system development, maintenance, and future expansion.

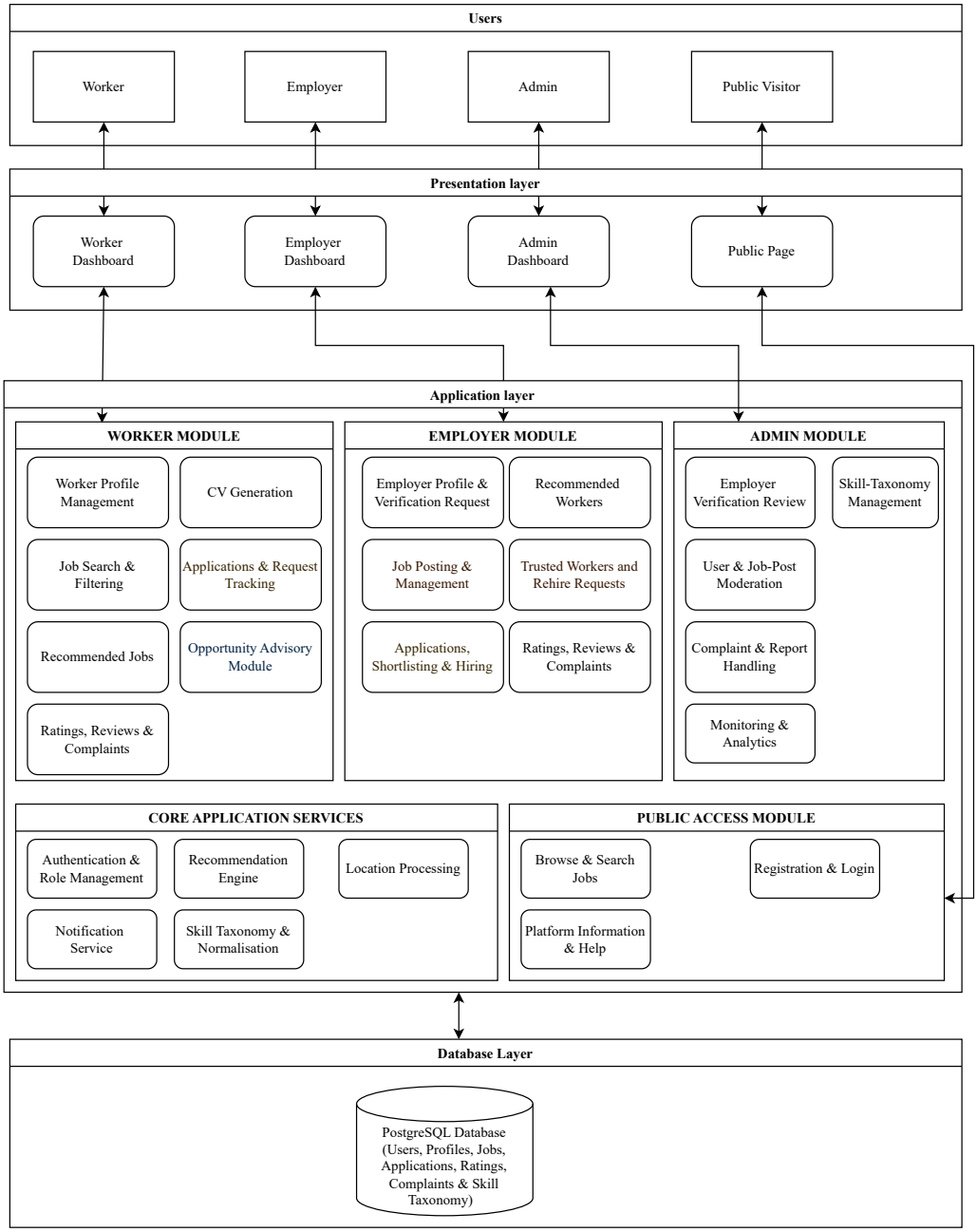


Figure 4.1: System Architecture

As shown in Figure 4.1, the platform follows a layered architecture consisting of the presentation, application, and database layers. At the top of the architecture, the main users are workers, employers, administrators, and public visitors. Each user interacts with the system through an appropriate interface in the presentation layer. Workers, employers, and administrators access their respective dashboards, while public visitors access the public page.

The application layer contains the main business logic of the platform. The worker module supports profile management, CV generation, job searching and filtering, application tracking, recommended jobs, opportunity advice, and the submission of ratings, reviews, and complaints. The employer module manages employer profiles and verification requests, job posting, worker recommendations, application shortlisting, hiring, ratings, reviews, complaints, and requests to rehire trusted workers. The admin module enables administrators to review employer verification requests, moderate users and job posts, manage the skill taxonomy, handle complaints and reports, and monitor platform analytics.

The application layer also includes shared core services, such as authentication and role management, the recommendation engine, location processing, notification services, and skill taxonomy normalisation. Public visitors can browse and search for jobs, register or log in, and access platform information and help through the public access module. The database layer uses PostgreSQL to store and manage the platform's data, including user accounts, profiles, jobs, applications, ratings, complaints, and the skill taxonomy. The arrows between the layers indicate the flow of requests and data throughout the system.

4.2. Operational Block Diagram and Feature Implementation

4.2.1. Operational Overview Diagram

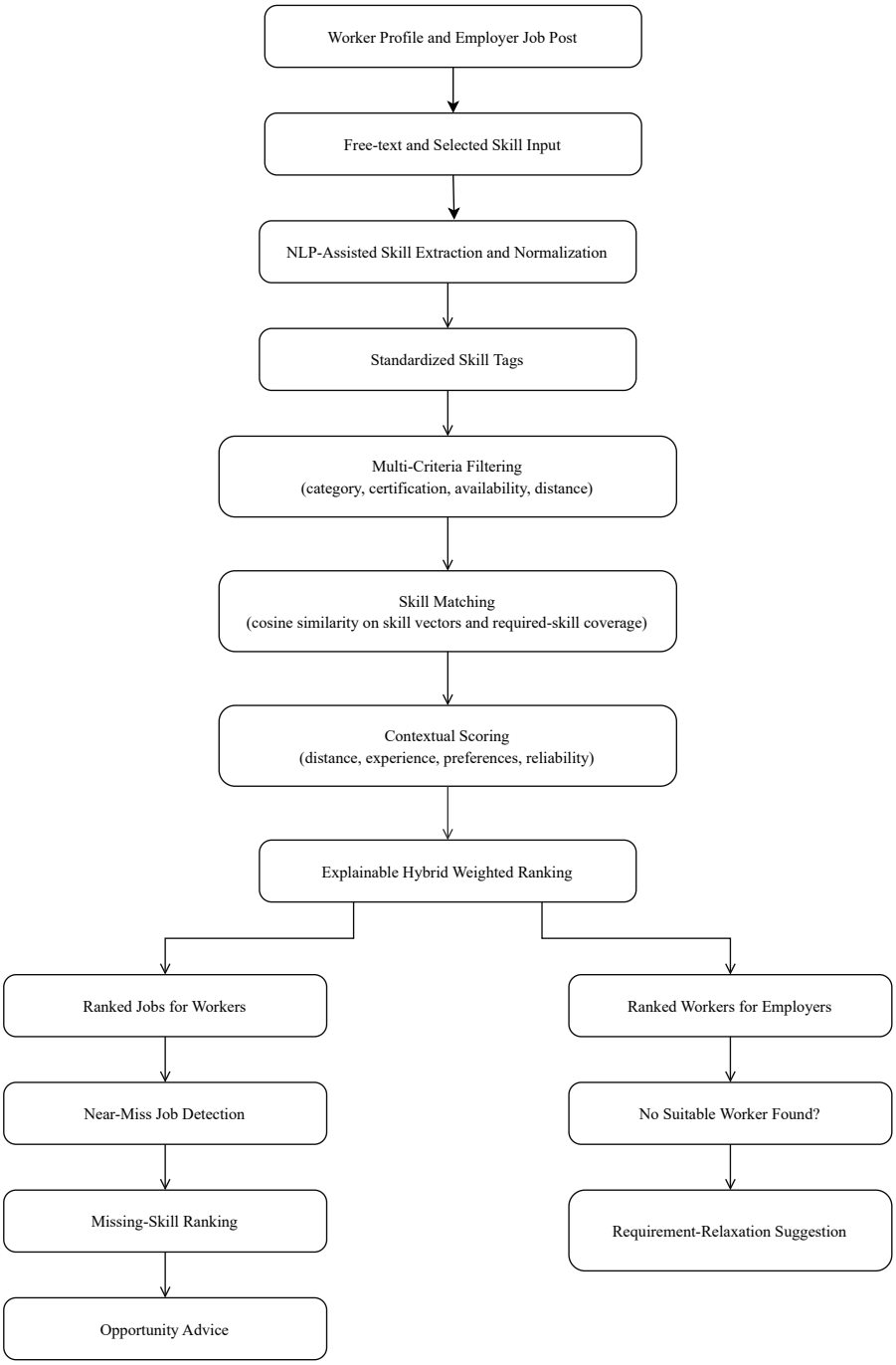


Figure 4.2: Operational flow of the Karmasheel recommendation and opportunity-advisory system

As shown in Figure 4.2, the system begins with two main inputs: the worker profile and the employer job post. The selected skills are first mapped to standardized skill records. Multi-criteria filtering then removes unsuitable matches based on category, certification, availability, and distance.

The filtered matches are evaluated using skill matching and contextual scoring. Cosine similarity will be used as a baseline for comparing standardized skill vectors, while required-skill coverage will measure the proportion of mandatory skills satisfied by a worker. Contextual scoring will consider distance, experience, preferences, and reliability indicators.

The remaining candidates are ranked using an explainable hybrid weighted scoring method. The resulting recommendations are presented in two directions: ranked jobs for workers and ranked workers for employers.

The worker-side branch supports opportunity advisory. Near-miss jobs are identified, missing skills are extracted and ranked, and practical improvement suggestions are shown to the worker. The employer-side branch supports requirement-relaxation suggestions when no suitable workers are found.

4.2.2. Structured Data Collection and Validation

The recommendation process depends on complete and consistent data. Therefore, the platform will collect worker profiles, employer profiles, and job-post details through structured forms with validation checks.

Worker profiles will contain contact information, address, location coordinates, category, subcategory, standardized skills, experience, availability, preferred job type, travel radius, wage expectation, and optional certificate records. Employer profiles will contain employer details, contact information, address, location coordinates, PAN or VAT number where applicable, verification status, and job-posting history. Job posts will contain job title, category, required skills, preferred skills, work location, schedule, duration, required experience, wage range, number of workers required, and application deadline.

The system will validate required fields before storing the data. This validation includes checking whether mandatory fields are filled, whether numerical values such as experience, wage range, and number of workers are valid, whether dates and schedules are

logically correct, whether selected skills belong to the standardized skill taxonomy, and whether location coordinates are available. Employer-submitted PAN or VAT details will be checked using rule-based format validation before being forwarded for admin review.

4.2.3. Structured Skill Taxonomy and NLP-Assisted Skill Normalization

Workers and employers may describe the same practical skill using different words. For example, a worker may use the phrase 'home wiring," while an employer may use the phrase 'residential electrical installation." Similarly, workers may enter mixed English and Romanized Nepali phrases such as 'ghar wiring," 'fan jodne," or "pipe leak banaune." If these phrases are stored as unrelated keywords, the system may overlook a suitable match.

To reduce this problem, Karmasheel will use a structured skill taxonomy. The taxonomy will organize skills into three levels:

- Category
- Subcategory
- Standardized Skill Tag

A category represents a broad area of work. A subcategory represents a more specific occupational area within that category. Each subcategory contains standardized skill tags that can be selected by workers and employers.

Table 4.1: Example of a Structured Skill Taxonomy

Category	**Subcategory**	**Standardized Skill Tags**
Construction and Repair	Electrical Work	House wiring, fan installation, switchboard installation, circuit repair, fault diagnosis
Construction and Repair	Plumbing	Pipe fitting, leakage repair, tap installation, bathroom-fixture maintenance
Hospitality and Services	Restaurant Services	Table service, food serving, order handling, customer assistance

Standardized Skill Tags

Each skill will be stored using one standard name. Workers will primarily select skills from the predefined list while creating or updating their profiles. Employers will also select required and preferred skills from the same taxonomy while creating job posts.

Using the same standardized records on both sides makes skill comparison more consistent. For example, the worker profile and the employer job post will both refer to the same database record for the skill "house wiring."

Skill Alias Mapping

The system will also maintain common synonyms, alternative phrases, and Romanized Nepali aliases for each standardized skill tag. This process is called NLP-assisted skill extraction and normalization. It helps the system map English and Romanized Nepali skill phrases to standardized skill records before matching is performed.

If a worker or employer enters an alternative phrase, the system will attempt to map that phrase to the corresponding standardized record.

Table 4.2: Example of English and Romanized Nepali Skill Alias Mapping

Entered Phrase or Alternative Term	**Standardized Skill Tag**
Residential wiring	House wiring
Home wiring	House wiring
House electrical installation	House wiring
Ghar wiring	House wiring
Ceiling-fan fitting	Fan installation
Fan jodne	Fan installation
Pipe leak banaune	Leakage repair
Plumbing leakage maintenance	Leakage repair
Dhara jodne	Tap installation

Text Preprocessing

Before mapping an entered phrase to a standardized skill tag, the system will apply simple text preprocessing steps. These steps may include lowercasing, punctuation removal, extra-space removal, tokenization, and spelling-variant handling. This helps the system compare user-entered English and Romanized Nepali phrases more consistently.

Confidence Score and Admin Review

After preprocessing, the system will match the entered phrase using alias lookup, synonym mapping, and fuzzy matching. If a high-confidence match is found, the phrase will be mapped automatically to the corresponding standardized skill tag, as shown in Figure 4.3.

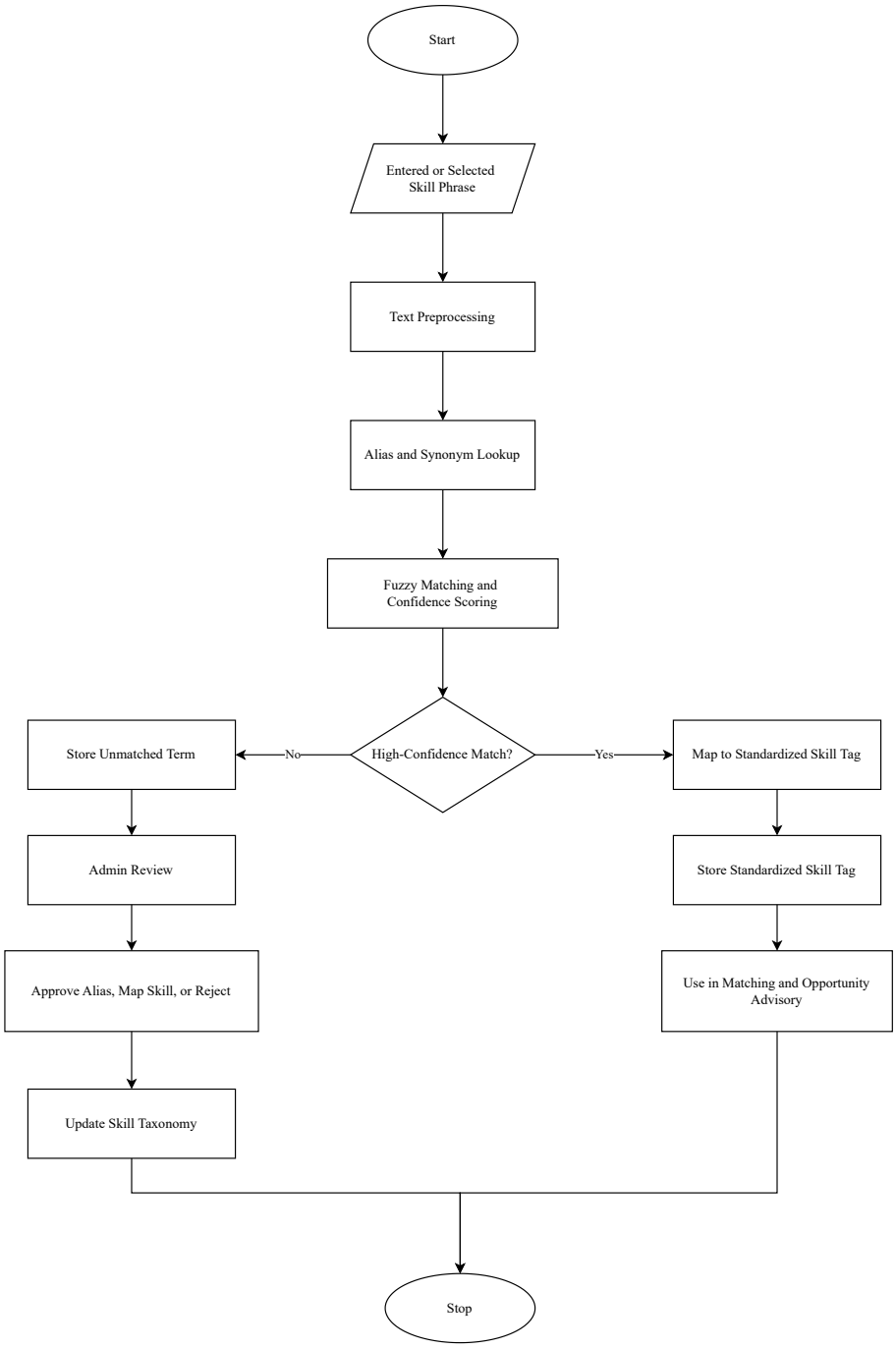


Figure 4.3: Normalization Flowchart

If the match confidence is low or no matching skill is found, the term will be stored for admin review instead of being added directly to the taxonomy. The admin may map the term to an existing skill, approve it as a new alias, or reject it. This prevents duplicate or incorrect skill records from being created automatically.

Evaluation Plan for Skill Normalization

The NLP-assisted skill-normalization module will be evaluated using a small manually labelled dataset of English and Romanized Nepali skill phrases. Each input phrase will have an expected standardized skill tag. The system output will be compared with the expected tag to check whether the phrase has been mapped correctly.

The evaluation will use the following simple measures:

- **Accuracy:** the percentage of input phrases correctly mapped to the expected standardized skill tag.
- **Unmatched rate:** the percentage of phrases for which the system does not find any suitable standardized skill tag.
- **Admin-review rate:** the percentage of phrases forwarded for admin review because of low confidence or missing taxonomy records.

Table 4.3: Example Dataset for Skill-Normalization Evaluation

Input Phrase	**Expected Standardized Skill Tag**
ghar wiring	House wiring
home wiring	House wiring
fan jodne	Fan installation
pipe leak banaune	Leakage repair
dhara jodne	Tap installation

This evaluation will help verify whether the normalization module can handle common English synonyms and Romanized Nepali skill phrases before they are used in skill matching and recommendation.

The structured skill taxonomy will support worker-profile creation, employer job posting, skill matching, automatic CV generation, opportunity-advisory suggestions, and admin analytics. The first prototype will use lightweight NLP-assisted methods such as text

preprocessing, synonym lookup, Romanized Nepali alias mapping, fuzzy matching, and admin review. More advanced semantic embeddings or trained NLP models may be considered as future enhancements after sufficient real-world data has been collected.

4.2.4. Employer Verification

Employer verification is included to improve trust and reduce misleading or fraudulent job posts. The first version of Karmasheel will use a semi-automated verification process. The system will perform rule-based validation of the submitted details, while the final verification decision will be made by the admin.

An employer profile may contain the organization or employer name, employer type, contact-person details, verified phone number, address and location, PAN or VAT number where applicable, supporting information or documents where required, and verification status.

Rule-Based PAN or VAT Format Validation

When an employer submits a PAN or VAT number, the system will perform basic rule-based checks before creating a verification request. These checks may include:

- checking whether the required field has been provided
- checking whether the value contains the expected number of digits
- checking whether the value contains only permitted characters
- checking whether the same identifier has already been submitted by another employer account
- checking whether the submitted contact information is complete

These checks do not prove that an employer is genuine. Their purpose is to detect incomplete, incorrectly formatted, or duplicate submissions before admin review.

Admin Review

After the submitted information passes the initial validation checks, the system will create a verification request for admin review. The admin will inspect the available employer details and compare them with accessible records where possible.

The admin may take one of the following actions:

- approve the verification request;
- reject the verification request; or
- request clarification or additional information.

The verification workflow is:

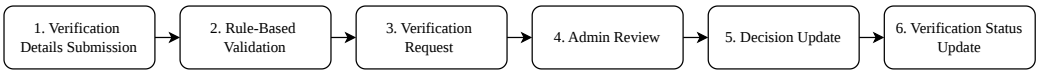


Figure 4.4: Verification workflow

The system will maintain a verification-status field. Possible values include:

Table 4.4: Employer Verification Statuses

Status	**Meaning**
Pending	The employer has submitted details and is waiting for admin review.
Verified	The admin has approved the request. The employer may access verified-employer features such as job posting and worker recommendations.
Rejected	The submitted details were not accepted.
Clarification Required	The admin has requested additional information or correction.

For household employers or local service requesters who may not have a PAN or VAT number, the platform may initially use phone verification, manual moderation, and restricted posting permissions.

The first prototype will not depend on automated access to government databases. More advanced verification integrations may be considered later if reliable and legally permitted services become available.

4.2.5. Job Search and Multi-Criteria Filtering

Workers will be able to search for available jobs using multiple criteria. Similarly, when an employer creates a job post, the system will retrieve workers who satisfy the most important conditions before applying detailed ranking. This process is called multi-criteria filtering.

The purpose of filtering is to reduce the number of clearly unsuitable results. The recommendation algorithm can then calculate detailed scores only for the remaining jobs or workers.

The filtering process will support both directions:

- **Worker-to-job filtering:** retrieve active job posts that are relevant to the worker's profile and preferences.
- **Job-to-worker filtering:** retrieve worker profiles that are relevant to the employer's job requirements.

Filtering Criteria

The main filtering criteria are summarized in Table 4.5.

Table 4.5: Multi-Criteria Filtering Rules

Criterion	**Purpose**	**Initial Treatment**
Category and subcategory	Retrieve jobs and workers from the relevant occupational area, such as electrical work or plumbing.	Hard filter
Availability	Check whether the worker is available during the required work date and time.	Hard filter
Required certification	Check whether the worker has the required certificate or training record when the job requires one.	Hard filter where applicable
Location radius	Remove results that are outside the acceptable travel radius. Distance may also influence the later ranking score.	Filter and ranking factor
Wage preference	Compare the offered wage with the worker's expected wage range.	Preference-based filter or scoring factor
Work-type preference	Compare the job type with the worker's preference, such as temporary, part-time, full-time, or shift-based work.	Preference-based filter or scoring factor

Some criteria will be treated as hard filters. For example, a worker who is not available on the required date should not be recommended for that job. Similarly, a worker without a mandatory certificate should be excluded when the certificate is essential.

Other criteria will be treated more flexibly. For example, a job slightly outside the preferred travel radius or wage range may receive a lower ranking score instead of being removed immediately. This allows the system to avoid discarding potentially useful opportunities.

Worker-to-Job Filtering Workflow

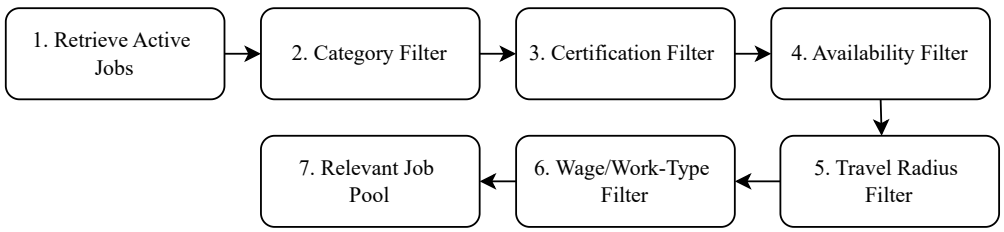


Figure 4.5: Worker-to-job filtering workflow

When a worker searches for jobs or requests recommendations, the platform will retrieve relevant active job posts and remove unsuitable jobs before detailed ranking is applied.

Job-to-Worker Filtering Workflow

When an employer creates a job post, the platform will retrieve relevant active worker profiles and remove unsuitable workers before detailed scoring and ranking are applied.

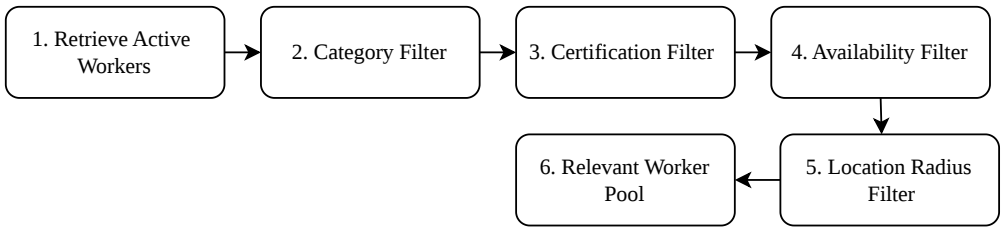


Figure 4.6: Job-to-worker filtering workflow

The filtered pools will then be passed to the hybrid recommendation module. That module will calculate skill similarity, distance scores, contextual scores, and the final explainable ranking.

4.2.6. Hybrid Recommendation Technique

After multi-criteria filtering has created a relevant candidate pool, the platform will calculate detailed match scores. Karmasheel will use a hybrid recommendation technique because a suitable match should not depend on a single factor. The method

combines standardized skill similarity, mandatory skill coverage, geographical distance, experience, reciprocal preferences, and reliability indicators.

The recommendation module will work in both directions:

- **Worker-to-job recommendation:** suitable jobs will be ranked for each worker.
- **Job-to-worker recommendation:** suitable workers will be ranked for each verified employer's job post.

Candidate Filtering

Before detailed scoring, the system will use the multi-criteria-filtering process described in the previous subsection. Category, availability, certification, and location conditions will be used to remove clearly unsuitable matches.

For normal recommendations, the platform will prioritize workers and jobs that satisfy the essential conditions. For opportunity-advisory analysis, partially matched jobs may also be retained so that near-miss opportunities can be identified.

Cold-Start Handling

The first version of Karmasheel may not have enough historical data such as previous applications, hiring outcomes, clicks, ratings, or completed jobs. Therefore, the recommendation module will reduce the cold-start problem by using structured worker-profile data and job-post data instead of depending on interaction history.

For new workers, the system will use selected skills, entered skill phrases, experience, location, availability, preferred job type, wage expectation, and travel radius. For new job posts, the system will use required skills, preferred skills, category, location, work schedule, wage range, and experience requirements. Skill taxonomy records and NLP-assisted normalization will help map different skill phrases to standardized skill tags before matching.

New workers without ratings will not be treated as unreliable. In the initial stage, reliability-related scoring will depend more on profile completeness, contact verification, and available verification indicators. Completed-job ratings and work history will gradually become more useful after real platform usage increases.

Cosine-Similarity Baseline

Cosine similarity will be used as a baseline method for comparing standardized worker-skill vectors and job-skill vectors. Each vector dimension represents a standardized skill in the taxonomy.

For example:

Worker skills: [house wiring, fan installation, fault diagnosis]

Job skills: [house wiring, fan installation, switchboard installation]

The cosine-similarity formula is:

$$\text{CosineSimilarity}(A, B) = \frac{A \cdot B}{\|A\|\|B\|} \tag{4.1}$$

where:

- A is the worker-skill vector;
- B is the job-skill vector;
- $A \cdot B$ is the dot product of the vectors; and
- $\|A\|$ and $\|B\|$ are the vector magnitudes.

The cosine-similarity value ranges from 0 to 1. A higher value indicates greater similarity between the worker's selected skills and the skills associated with the job post.

For easier interpretation, the value may be converted into a percentage:

$$\text{CosineSimilarityScore} = \text{CosineSimilarity}(A, B) \times 100 \tag{4.2}$$

Cosine similarity will be used as a baseline rather than the only ranking method because it does not fully represent mandatory requirements, distance, availability, or preferences.

Required-Skill Coverage

The platform will calculate the percentage of mandatory skills satisfied by a worker. This is important because required skills should receive greater importance than preferred skills.

$$\text{RequiredSkillCoverage} = \frac{\text{Number of matched required skills}}{\text{Total number of required skills}} \times 100 \tag{4.3}$$

For example, if a worker matches two out of three required skills:

$$\text{RequiredSkillCoverage} = \frac{2}{3} \times 100 = 66.67\% \tag{4.4}$$

A provisional structured skill score can be calculated as:

$$\text{SkillScore} = w_{\text{req}}(\text{RequiredSkillCoverage}) + w_{\cos}(\text{CosineSimilarityScore}) \tag{4.5}$$

where w_{req} and $w_{\cos}$ are configurable weighting coefficients for required-skill coverage and cosine-similarity score, respectively. The weights satisfy:

$$w_{\text{req}} + w_{\cos} = 1$$

For the initial prototype, $w_{\text{req}} = 0.70$ and $w_{\cos} = 0.30$ will be used.

This gives greater importance to mandatory skills while still using cosine similarity as a baseline measure of overall skill-vector similarity. Preferred skills may be used as an additional ranking bonus. The exact weights may be adjusted after testing.

Haversine-Distance Scoring

Location is particularly important for local, temporary, and urgent jobs. The platform will calculate the geographical distance between the worker and the workplace using latitude and longitude.

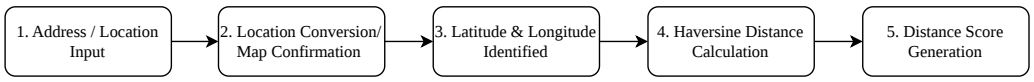


Figure 4.7: Location-Based Distance Scoring Workflow

The Haversine-distance formula is:

$$d = 2R \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)}\right) \tag{4.6}$$

where:

- d is the geographical distance
- R is the radius of the Earth
- ϕ_1 and ϕ_2 are the latitudes of the worker and workplace
- λ_1 and λ_2 are their longitudes
- $\Delta\phi = \phi_2 - \phi_1$ and
- $\Delta\lambda = \lambda_2 - \lambda_1$.

The calculated distance will be converted into a normalized distance score. For the initial prototype, the maximum acceptable distance is set to $D_{\max} = 20$ km. The distance score is calculated as:

$$\text{DistanceScore} = \begin{cases} \left(1 - \dfrac{d}{D_{\max}}\right) \times 100, & \text{if } d \leq D_{\max} \\ 0, & \text{if } d > D_{\max} \end{cases} \tag{4.7}$$

where d is the Haversine distance between the worker and the workplace, and $D_{\max}$ is the maximum acceptable distance. For the initial prototype, $D_{\max}$ is set to 20 km, but this value may be adjusted according to the job type, worker travel preference, and local travel conditions.

This scoring method gives nearby workers or jobs higher scores and gradually reduces the score as the distance increases. For urgent or one-day jobs, distance may receive greater importance in the final weighted ranking.

Definition of Supporting Sub-Scores

In addition to skill and distance scores, the final ranking will use supporting sub-scores. These sub-scores convert practical matching factors into values between 0 and 100 so that they can be combined in the final weighted ranking.

- **ExperienceScore:** This score compares the worker's experience with the experience expected in the job post. If the worker meets or exceeds the required experience, the score may be set to 100. If the worker has less experience, the score may be reduced proportionally.
- **AvailabilityScore:** This score checks whether the worker is available during the required work date and time. If availability is a mandatory condition, unavailable workers will be removed during filtering. If it is flexible, availability may be used as a scoring factor.
- **WagePreferenceScore:** This score compares the offered wage with the worker's expected wage range. Jobs within the expected range will receive a higher score, while jobs outside the range may receive a lower score instead of being removed immediately.
- **WorkTypeScore:** This score compares the job type with the worker's preference, such as temporary, part-time, full-time, shift-based, or urgent work.
- **ReliabilityVerificationScore:** This score considers trust indicators such as contact verification, employer verification, profile completeness, completed-job records, and ratings. For new users, the score will not depend heavily on ratings because they may not have completed jobs yet.

These sub-scores will be implemented using simple rule-based mappings in the first prototype. The mappings will remain configurable so that they can be adjusted after testing with sample worker profiles and job posts.

Experience and Reliability-Verification Score Calculation

The experience score will compare the worker's experience with the experience required in the job post. If the worker meets or exceeds the required experience, the score will be set to 100. If the worker has less experience, the score will be reduced proportionally.

$$\text{ExperienceScore} = \min\left(\frac{\text{WorkerExperience}}{\text{RequiredExperience}}, 1\right) \times 100 \tag{4.8}$$

If the job post does not specify required experience, the experience score may be treated as a neutral or full score so that the worker is not penalized for a requirement that the employer has not provided.

The reliability-verification score will be calculated from trust-related indicators such as contact verification, profile completeness, completed-job records, rating records, and complaint history. Since new users may not have completed jobs or ratings, the score will not depend heavily on rating history during the initial stage.

Reliability-Verification Score Calculation

General Reliability-Verification Formula In general, the reliability-verification score can be represented as a weighted combination of rating score, completed-job score, clean record score, and verification status score.

$$\begin{aligned}\text{ReliabilityVerificationScore} = {} & w_{\text{rating}}T + w_{\text{job}}J \\ & + w_{\text{record}}R + w_{\text{verify}}V\end{aligned} \tag{4.9}$$

where w_{rating}, w_{job}, w_{record}, and w_{verify} are configurable weighting coefficients for rating score, completed-job score, clean record score, and verification status score, respectively. These weights satisfy:

$$w_{\text{rating}} + w_{\text{job}} + w_{\text{record}} + w_{\text{verify}} = 1 \tag{4.10}$$

The reliability-verification score will measure the trustworthiness and work history of a worker or employer. Since new users may not have completed jobs or ratings, the system will not use rating and completed-job history until a minimum amount of valid work history is available. This prevents the cold-start problem and avoids ranking new genuine users unfairly low.

Rating Score Let $N_{\min}$ be the minimum number of valid completed-job ratings required before rating and completed-job history are included in the reliability score. For the initial prototype, $N_{\min} = 3$ will be used.

$$N_{\min} = 3 \tag{4.11}$$

The rating score will be calculated only from valid ratings linked to completed jobs. Ratings not linked to completed work will not be used. To avoid unfairly ranking new users, the rating score will be included in the reliability- verification score only when the user has received at least $N_{\min}$ valid ratings.

The rating score will be used only when:

$$N_{\text{ratings}} \geq N_{\min} \tag{4.12}$$

When this condition is satisfied, the rating score will be calculated as:

$$T = \left(\frac{\text{AverageRating} - 1}{4} \right) \times 100 \tag{4.13}$$

where:

- T is the rating score;
- AverageRating is the average rating on a scale of 1 to 5;
- N_{ratings} is the number of valid ratings received; and
- $N_{\min}$ is the minimum number of valid ratings required before the rating score is included.

The formula converts the average rating from a 1–5 scale into a 0–100 score. For example, an average rating of 5 becomes 100, while an average rating of 1 becomes 0. If $N_{\text{ratings}} < N_{\min}$, the rating score will not be included in the reliability-verification score.

Completed-Job Score The completed-job score represents the user's completed work history. A higher number of completed jobs indicates more platform-based work experience. For the initial prototype, the score will reach 100 after $J_{\max} = 10$ completed jobs.

$$J = \min\left(\frac{N_{\text{completed}}}{J_{\max}}, 1\right) \times 100 \tag{4.14}$$

where:

- J is the completed-job score;
- $N_{\text{completed}}$ is the number of valid completed jobs; and
- $J_{\max}$ is the completed-job count required to receive the full score.

For the initial prototype, $J_{\max} = 10$ will be used. This value may be adjusted after testing.

Clean Record Score The clean record score checks whether the user has valid complaints, warnings, or restrictions recorded by the system. Only admin-reviewed complaints will be used so that unverified reports do not unfairly reduce the score.

$$R = \max\left(0, 100 - 25N_{\text{complaints}} - 10N_{\text{warnings}} - 50N_{\text{restrictions}}\right) \tag{4.15}$$

where:

- $N_{\text{complaints}}$ is the number of valid admin-reviewed complaints;
- N_{warnings} is the number of admin warnings; and
- $N_{\text{restrictions}}$ is the number of active account restrictions.

A new user with no valid complaints, warnings, or restrictions will receive a clean record score of 100.

Verification Status Score The verification status score represents the current verification state of a worker or employer. For workers, it may refer to profile or certificate verification where applicable. For employers, it refers to employer verification.

$$V = \begin{cases} 100, & \text{if verification is approved or not required} \\ 50, & \text{if verification has been submitted and is pending} \\ 0, & \text{if verification is required but missing or rejected} \end{cases} \tag{4.16}$$

Final Reliability-Verification Score If the user does not have enough valid ratings, the system will use a cold-start reliability score based on clean record and verification status. This avoids unfairly penalizing new users.

$$\text{ReliabilityVerificationScore} = 0.70R + 0.30V \tag{4.17}$$

This formula will be used when:

$$N_{\text{ratings}} < N_{\text{min}} \tag{4.18}$$

After the user has at least N_{min} valid completed-job ratings, the system will give higher priority to rating score and completed-job score.

$$\text{ReliabilityVerificationScore} = 0.45T + 0.30J + 0.15R + 0.10V \tag{4.19}$$

This formula will be used when:

$$N_{\text{ratings}} \geq N_{\text{min}} \tag{4.20}$$

where:

- T is the rating score;
- J is the completed-job score;
- R is the clean record score; and
- V is the verification status score.

This approach gives priority to real work history after enough valid records are available, while still protecting new users from being ranked unfairly low. The weights, $N_{\min}$, and $J_{\max}$ are provisional and may be adjusted after testing with sample worker profiles, job posts, completed-job records, and feedback data.

Reciprocal Preference Scoring

A suitable job match should satisfy both the employer and the worker. The platform will therefore use reciprocal preference scoring.
Employer-side suitability may consider:

- required and preferred skills;
- experience;
- certification;
- availability; and
- reliability indicators.

Worker-side suitability may consider:

- expected wage range;
- preferred travel distance;
- work type;
- work schedule;
- preferred category; and
- employer-verification status.

A possible initial formula is:

$$\text{ReciprocalPreferenceScore} = w_{\text{emp}}(\text{EmployerSideSuitability}) + w_{\text{worker}}(\text{WorkerSideSuitability}) \tag{4.21}$$

where w_{emp} and w_{worker} are configurable weighting coefficients for employer-side suitability and worker-side suitability, respectively. The weights satisfy:

$$w_{\text{emp}} + w_{\text{worker}} = 1$$

For the initial prototype, $w_{\text{emp}} = 0.60$ and $w_{\text{worker}} = 0.40$ will be used.

The weights are provisional. They may be changed after testing to achieve a more balanced result.

Weighted Final Ranking

After calculating the individual scores, the system will combine them into an explainable final match score.

A possible initial formula is:

$$\begin{aligned} \text{FinalMatchScore} = {} & w_{\text{skill}}(\text{SkillScore}) \\ & + w_{\text{dist}}(\text{DistanceScore}) \\ & + w_{\text{exp}}(\text{ExperienceScore}) \\ & + w_{\text{pref}}(\text{ReciprocalPreferenceScore}) \\ & + w_{\text{rel}}(\text{ReliabilityVerificationScore}) \end{aligned} \tag{4.22}$$

where w_{skill}, w_{dist}, w_{exp}, w_{pref}, and w_{rel} are configurable weighting coefficients for skill, distance, experience, reciprocal preference, and reliability-verification scores, respectively. The weights satisfy:

$$w_{\text{skill}} + w_{\text{dist}} + w_{\text{exp}} + w_{\text{pref}} + w_{\text{rel}} = 1$$

For the initial prototype, the values shown in Table 4.6 will be used.

Table 4.6 summarizes the initial weighting scheme.

Table 4.6: Provisional Components of the Final Match Score

Component	**Initial Weight**
Skill score	40%
Distance score	20%
Experience score	15%
Reciprocal-preference score	15%
Reliability and verification score	10%

The final match score will be used to sort the filtered results from highest to lowest suitability. The exact weights and thresholds will remain configurable and may be adjusted after testing with sample worker profiles and job posts.

Weight Selection and Evaluation Plan

The initial weights used in the hybrid ranking formula are heuristic values based on the expected importance of each matching factor. Skill matching is given the highest priority because the worker must be capable of performing the required work. Distance, experience, reciprocal preferences, and reliability indicators are then used to refine the ranking.

The weights will be evaluated using a small test dataset containing sample worker profiles and job posts. Each worker-job pair may be manually labelled as suitable, near-miss, or unsuitable. The recommendation output will then be compared with these expected labels.

The evaluation will use simple measures such as:

- **Precision@5:** The proportion of the top five recommended jobs or workers that are actually suitable.
- **Near-miss accuracy:** Whether the opportunity-advisory module correctly identifies realistic near-miss jobs.
- **Coverage:** Whether active jobs receive at least one suitable worker recommendation and whether workers receive at least one relevant job recommendation.
- **Manual review:** Project supervisor, team, or domain-review feedback on whether the recommended results are reasonable.

Table 4.7: Evaluation Plan for Weight Adjustment

Component	**Reason for Priority**	**Evaluation Method**
Skill score	Worker must satisfy the practical work requirements.	Check whether top recommendations satisfy required skills.
Distance score	Local jobs depend strongly on travel practicality.	Check whether recommended jobs or workers are within an acceptable distance.
Experience score	Experience improves suitability for practical work.	Compare ranking order with manually labelled suitable matches.
Reciprocal-preference score	The match should be acceptable to both worker and employer.	Check wage, schedule, work type, and travel preferences.
Reliability-verification score	Trust indicators are important in local hiring.	Review the effect of verification and completed-job records on ranking.

Based on the evaluation results, the weights and thresholds may be adjusted before finalizing the prototype. This prevents the weighting scheme from being treated as a fixed assumption.

Explainable Match Reasons

The platform will show understandable reasons with each recommendation instead of displaying only a numerical score. This improves transparency and helps workers and employers understand the recommendation.

An employer-side recommendation may be displayed as follows:

Recommended Worker: Ram Bahadur

Overall Match: 86%

Reasons:

- Matches four out of five required skills.
- Located 2.9 km from the workplace.
- Available on the required date.
- Has four years of relevant experience.

- Phone-verified profile.

A worker-side recommendation may be displayed as follows:

Recommended Job: Electrician Required in Patan
Overall Match: 84%
Reasons:

- Matches four out of five required skills.
- Workplace is 3.1 km away.
- Job date matches your availability.
- Offered wage is within your preferred range.
- Employer profile is verified.

The recommendation module will therefore combine transparent formulas with clear textual explanations. This makes the first prototype practical, testable, and easier to improve after evaluation.

4.2.7. Opportunity-Advisory Feature

The opportunity-advisory feature extends the recommendation module by helping workers understand why some jobs are almost suitable but not fully reachable. A normal recommendation answers which jobs are currently suitable for a worker. The opportunity-advisory feature answers which practical skills may help the worker qualify for more relevant jobs in the future.

Near-Miss Job Detection

A near-miss job is a job for which the worker already satisfies a meaningful portion of the requirements but misses one or more important skills or conditions. The system will first retrieve jobs from relevant categories and then calculate the worker–job match score. Jobs that are not strong enough to be recommended but are still practically close to the worker's profile will be treated as near-miss jobs.
A provisional interpretation is:

Table 4.8: Provisional Near-Miss Job Interpretation

Match Score Range	**Interpretation**
Below 40%	Not currently relevant for opportunity advice.
40%–75%	Near-miss job that may be useful for skill-gap analysis.
Above 75%	Strong recommendation that may be shown as a suitable job.

The thresholds will remain configurable. The system may also consider the number and importance of missing required skills because a worker missing many essential skills should not be treated as a near-miss case only because of a moderate numerical score.

Missing-Skill Identification

After near-miss jobs are identified, the system will compare each near-miss job's required and preferred skills with the worker's standardized skill set. Skills present in the job post but absent from the worker profile will be listed as missing skills.

Required skills will receive higher importance than preferred skills because they directly affect eligibility. Preferred skills may still be considered because they can improve the worker's ranking for future opportunities.

Missing-Skill Ranking

The system will rank missing skills according to their opportunity impact. For the first prototype, the priority score will be calculated from near-miss jobs only. A missing skill will receive a higher score if it appears in many near-miss jobs, appears as a required skill, belongs to nearby jobs, and is related to the worker's selected or preferred work category.

Let w be a worker and let $\mathcal{N}_w$ be the set of near-miss jobs identified for that worker. For a missing skill s, let $\mathcal{J}_{w,s}$ be the subset of near-miss jobs in which skill s is required or preferred by the job but missing from the worker profile:

$$\mathcal{J}_{w,s} = \left\{ j \in \mathcal{N}_w : s \in (R_j \cup P_j) \text{ and } s \notin S_w \right\}$$

where S_w is the set of standardized skills already possessed by worker w, R_j is the set of required skills for job j, and P_j is the set of preferred skills for job j. If $\mathcal{N}_w$ or $\mathcal{J}_{w,s}$

is empty, the system will not calculate a priority score for that skill.

For the initial prototype, the skill-importance value for skill s in job j is defined as:

$$\mathrm{Imp}_{j,s} = \begin{cases} 1.00, & \text{if } s \in R_j, \\ 0.50, & \text{if } s \in P_j \text{ and } s \notin R_j, \\ 0, & \text{otherwise.} \end{cases}$$

The location relevance of job j for worker w is calculated using the distance between the worker and the job location:

$$\mathrm{Loc}_{w,j} = \max\left(0, 1 - \frac{d_{w,j}}{D_w}\right)$$

where $d_{w,j}$ is the Haversine distance between worker w and job j in kilometers. D_w is the maximum nearby distance considered for the worker. In the first prototype, D_w will be the worker's preferred travel radius if available; otherwise, a default value of 20 km will be used.

The category relevance of job j for worker w is defined as:

$$\mathrm{Cat}_{w,j} = \begin{cases} 1.00, & \text{if } j \text{ belongs to the worker's selected subcategory,} \\ 0.80, & \text{if } j \text{ belongs to the worker's selected main category,} \\ 0.60, & \text{if } j \text{ belongs to a preferred work category,} \\ 0.30, & \text{otherwise.} \end{cases}$$

Using these values, the four components of the missing-skill priority score are calculated as:

$$\begin{aligned} \mathrm{Frequency}(s) &= \frac{|\mathcal{J}_{w,s}|}{|\mathcal{N}_w|}, \\ \mathrm{SkillImportance}(s) &= \frac{1}{|\mathcal{J}_{w,s}|} \sum_{j \in \mathcal{J}_{w,s}} \mathrm{Imp}_{j,s}, \\ \mathrm{LocationRelevance}(s) &= \frac{1}{|\mathcal{J}_{w,s}|} \sum_{j \in \mathcal{J}_{w,s}} \mathrm{Loc}_{w,j}, \\ \mathrm{CategoryRelevance}(s) &= \frac{1}{|\mathcal{J}_{w,s}|} \sum_{j \in \mathcal{J}_{w,s}} \mathrm{Cat}_{w,j}\,. \end{aligned} \tag{4.23}$$

The final missing-skill priority score is then calculated as:

$$\text{SkillPriorityScore}(s) = \text{Frequency}(s) \times \text{SkillImportance}(s) \times \text{LocationRelevance}(s) \times \text{CategoryRele} \tag{4.24}$$

where:

- **Frequency** measures the proportion of near-miss jobs in which the missing skill appears.
- **SkillImportance** gives higher value to required skills than preferred skills.
- **LocationRelevance** gives higher value to missing skills that appear in nearby job opportunities.
- **CategoryRelevance** gives higher value to skills related to the worker's selected category, subcategory, or preferred work area.

All component values are normalized between 0 and 1. Therefore, the final skill-priority score will also lie between 0 and 1. The system will use this score to rank missing skills from highest to lowest priority.

For example, suppose the missing skill "fault diagnosis" appears in three out of five near-miss electrical jobs. It is required in two jobs and preferred in one job. If the location relevance values are 0.80, 0.70, and 0.60, and all three jobs belong to the worker's selected category, then:

$$\text{Frequency} = \frac{3}{5} = 0.60$$

$$\text{SkillImportance} = \frac{1.00 + 1.00 + 0.50}{3} \approx 0.83$$

$$\text{LocationRelevance} = \frac{0.80 + 0.70 + 0.60}{3} = 0.70$$

$$\text{CategoryRelevance} = 1.00$$

$$\text{SkillPriorityScore} = 0.60 \times 0.83 \times 0.70 \times 1.00 \approx 0.35$$

The system will compare this value with the priority scores of other missing skills. If "fault diagnosis" receives one of the highest scores, it will be shown as a high-priority skill-improvement suggestion for the worker.

Practical Skill-Improvement Suggestions

The opportunity-advisory output will be shown in an understandable format rather than only as a number. For example, the system may display:

> **Suggested Skill:** Fault diagnosis **Reason:** You already meet most requirements for several nearby electrical-work jobs, but fault diagnosis is commonly missing from your profile. Adding this skill may help you qualify for more similar jobs.

The system will not guarantee that learning a suggested skill will result in employment. It will only identify practical skill gaps based on currently available near-miss jobs and worker-profile data.

Employer-Side Requirement-Relaxation Suggestions

The opportunity-advisory logic will also support employers when no suitable worker is found for a job post. In such cases, the system may test whether moving a less essential required skill to the preferred-skill list increases the number of eligible workers.
For example, if no worker satisfies all required skills for an electrical job, the system may suggest:

- No workers currently match all required skills. Consider moving "switchboard installation" from required to preferred. This may allow more workers to become eligible for review.

The employer will remain responsible for deciding whether the suggestion is acceptable. The system will only provide a recommendation and will not automatically change job requirements.

4.2.8. Application and Status Tracking

Workers may apply for jobs discovered through search results or recommendations. Employers will be able to review submitted applications and update their statuses. The

platform will use a state-based application workflow so that an application can move only through permitted transitions.

Viewed and saved jobs may be tracked separately as user activities. Once a worker submits an application, the application-status field will initially be set to ”Applied”.

Table 4.9 summarizes the main application statuses.

Table 4.9: Application Statuses

Status	**Meaning**
Applied	The worker has submitted an application for the job.
Shortlisted	The employer has selected the worker for further consideration.
Contacted	The employer has contacted the worker for additional discussion or confirmation.
Hired	The worker has been selected for the job.
Completed	The job has been completed and may now receive ratings or feedback.
Rejected	The employer has decided not to select the worker.
Withdrawn	The worker has withdrawn the application.
Cancelled	The job post or application process has been cancelled.

The basic workflow is:

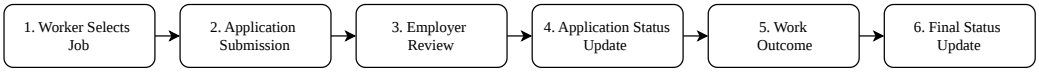


Figure 4.8: Application-status tracking workflow

The system will validate each transition before updating the stored status. For example, a rating should not be submitted before a job has been marked as completed.

The platform will also support direct requests. A verified employer may send a job request to a recommended worker. The worker may accept or reject the request. An accepted request can proceed to the hiring and completion stages.

4.2.9. Automatic CV Generation

Many blue-collar workers may not already have a formal CV. Karmasheel will generate a simple CV automatically using the information stored in the worker profile.

The generated CV may contain:

- Worker name
- verified contact number
- location
- professional summary
- selected skills
- work experience
- preferred work type
- availability
- certification or training details, where applicable and
- verification status.

Template-Based Document Generation

The first version will use a predefined CV template. Worker-profile data will be inserted into the appropriate sections of the template automatically. The professional summary will initially be generated using simple rules rather than a complex language model. This approach is practical for the first prototype because it produces a consistent CV without requiring the worker to understand formal document-writing conventions.

4.2.10. Ratings, Reviews, and Rehire Support

After a job has been marked as completed, the worker and employer may submit ratings and reviews for each other. Ratings provide a simple reliability indicator that may support future recommendations and rehiring decisions.

Rating Aggregation

The platform will calculate the average rating received by a worker or employer using completed-job reviews.

$$\text{AverageRating} = \frac{\text{Sum of received ratings}}{\text{Number of valid work-engagement reviews}} \tag{4.25}$$

The average rating will be calculated only when at least one valid completed-job review exists. Reviews that are not linked to a completed job will not be included.

The rating flowchart is:

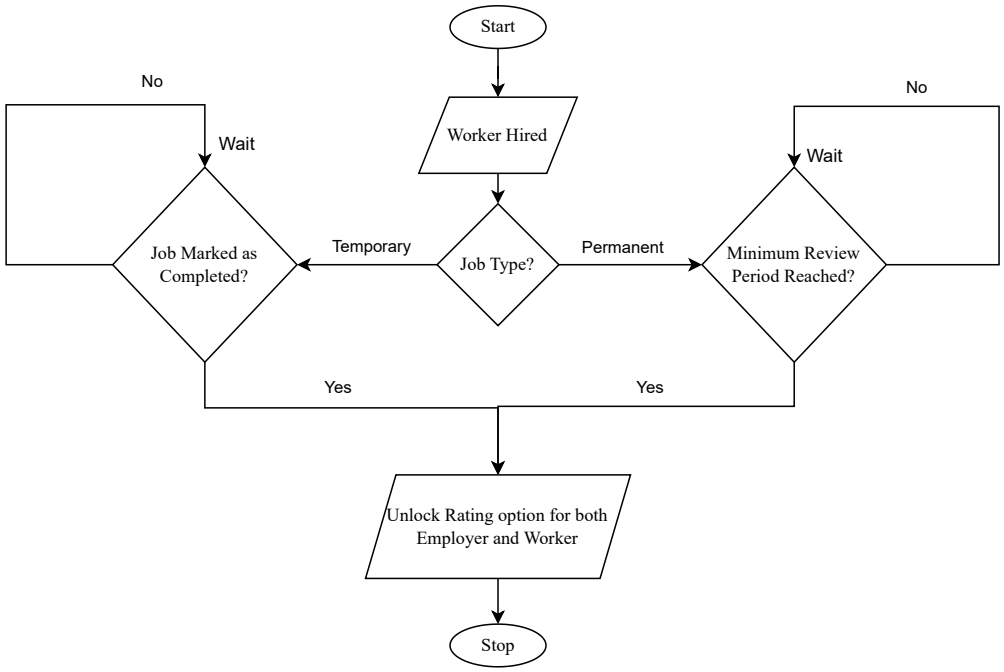


Figure 4.9: Rating workflow for temporary and permanent jobs in the Karmasheel platform

Trusted-Worker Filtering and Rehiring

An employer may save a reliable worker to a trusted-worker list after a successful job. The saved worker can later be considered for similar opportunities.

The workflow is:

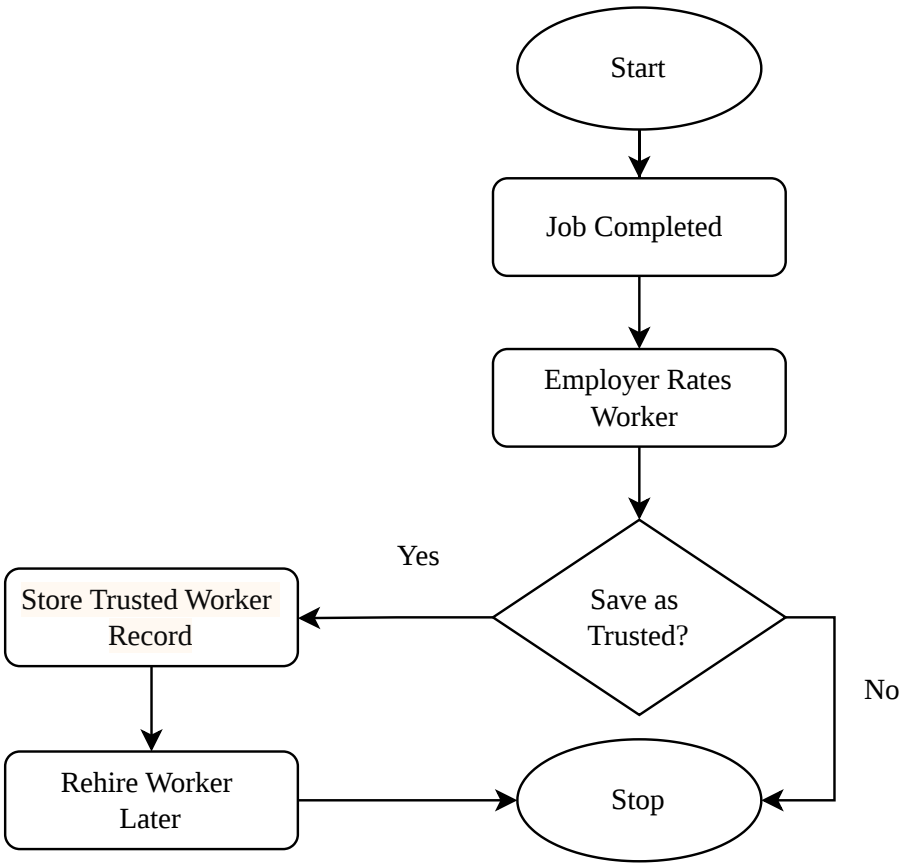


Figure 4.10: Trusted-Worker Filtering and Rehiring

The trusted-worker list will not replace the recommendation algorithm. It will provide an additional convenience for employers who want to rehire workers with whom they have already had a satisfactory experience.

4.2.11. Complaint Handling and Admin Monitoring

Workers and employers will be able to report suspicious activity or problems encountered through the platform. The first prototype will use rule-based flagging followed by manual admin moderation.

A complaint may relate to:

- fake or misleading job posts
- incorrect profile information
- payment-related issues

- suspicious employer or worker activity
- inappropriate behavior or
- misuse of the platform.

Rule-Based Flagging

Each complaint will be stored with a category, description, linked user or job post, submission date, and review status. The system will count complaints associated with the relevant account or job post.

The workflow is:

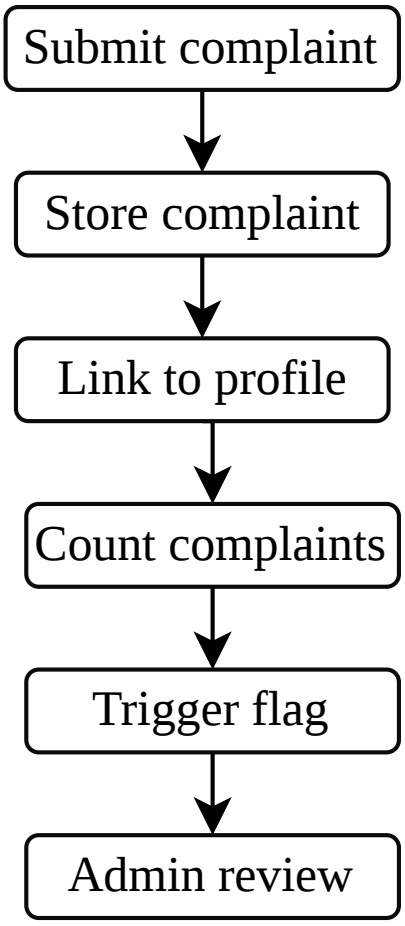


Figure 4.11: Rule based flagging workflow

The initial flagging threshold will remain configurable. A flag does not prove that a user has violated platform rules. It only indicates that admin review is required.

Manual Moderation

After reviewing a complaint or flagged record, the admin may:

- mark the complaint as reviewed
- request clarification
- issue a warning
- restrict an account
- suspend an account
- remove inappropriate content or
- correct inaccurate information.

The final decision will remain with the admin during the first version of the platform.

Usage Analytics and Recommendation Monitoring

The admin dashboard will display basic metrics for monitoring platform usage and recommendation quality. Possible metrics include:

- total registered workers
- total registered and verified employers
- total and active job posts
- recommendation-view rate
- application rate
- shortlist rate
- hire rate
- number of completed jobs
- average recommended distance
- most active job categories
- jobs receiving no suitable-worker recommendations
- workers receiving no suitable-job recommendations

- most frequently missing skills
- number of complaints and flagged records.

These metrics will help the team identify practical issues, such as categories with insufficient worker profiles, overly strict job requirements, and frequently missing skills. More advanced analytics may be added after the platform has collected sufficient real-world data.

4.3. Incremental Model

The proposed development increments are as follows:

1. **Increment 1: User registration and authentication:** This increment will include worker and employer registration, contact verification, login, logout, password security, and role-based access control.
2. **Increment 2: Profile, taxonomy, and NLP-assisted skill management:** This increment will include worker-profile management, employer-profile management, skill categories, standardized skill tags, synonym mappings, Romanized Nepali aliases, availability details, work preferences, and employer-verification requests. It will also include the basic NLP-assisted skill normalization workflow for mapping entered skill phrases to standardized skill tags.
3. **Increment 3: Job posting and application management:** This increment will allow verified employers to create and manage job posts. Skills entered in job posts will be processed using the skill taxonomy and NLP-assisted normalization before being stored as required and preferred skills. Workers will be able to search for jobs, apply filters, submit applications, and track application statuses.
4. **Increment 4: Recommendation and matching module:** This increment will provide worker-to-job and job-to-worker recommendations. Candidate filtering and explainable weighted ranking will consider normalized skills, location, availability, experience, preferences, and reliability indicators. The standardized skill tags produced by the normalization process will be used to create comparable worker-skill and job-skill records for matching.

5. **Increment 5: Supporting worker and employer features:** This increment will include automatic CV generation, notifications, direct worker requests, trusted-worker rehiring, job-completion confirmation, ratings, and reviews.

6. **Increment 6: Opportunity advisory and administration:** This increment will add near-miss job analysis, skill-gap suggestions, complaint handling, flagged-content moderation, analytics, and the admin dashboard. The admin dashboard will also support review of unmatched or low-confidence skill terms so that the admin can approve aliases, map terms to existing skill tags, or reject invalid entries.

7. **Increment 7: Integration, testing, and refinement:** The final increment will focus on integrating the modules, correcting identified issues, improving usability, and verifying that the complete platform satisfies its functional and non-functional requirements. It will also test the skill-normalization workflow using sample English and Romanized Nepali skill phrases to check whether common skill variations are correctly mapped to standardized skill tags.

4.4. Tools Used

The following tools and technologies will be used during the development:

- **Python, Django, and Django REST Framework:** Python will be used as the primary programming language, Django will be used as the backend web framework, and Django REST Framework will be used to develop APIs. Together, they will support authentication, role-based access, profile management, job posting, recommendation logic, application tracking, complaint handling, and communication between the frontend and backend.

- **PostgreSQL:** PostgreSQL will be used as the relational database-management system. It will store user accounts, worker profiles, employer profiles, skill taxonomy records, job posts, applications, recommendations, verification requests, ratings, complaints, notifications, and skill-normalization records.

- **HTML, CSS, and JavaScript:** HTML, CSS, and JavaScript will be used to build the frontend interface, including page structure, styling, form validation, filtering, and basic interactive behavior.

- **Regular Expressions and RapidFuzz:** Regular expressions and RapidFuzz will be used for lightweight NLP-assisted skill extraction and normalization. They will support text preprocessing, alias matching, fuzzy matching, and mapping of common English and Romanized Nepali skill phrases to standardized skill tags.
- **HTML-to-PDF Generation Tool:** An HTML-to-PDF generation tool such as WeasyPrint or xhtml2pdf will be used to generate downloadable CVs from pre-defined HTML templates filled with worker-profile information.
- **Git and GitHub:** Git will be used for version control, while GitHub will be used to store the source code, track changes, manage collaboration, and maintain the development history of the project.
- **Postman:** Postman will be used to test the APIs developed using Django REST Framework. It will help verify request handling, response formats, authentication, authorization, and error conditions.
- **Draw.io:** Draw.io will be used to prepare system-design diagrams such as use-case diagrams, data-flow diagrams, workflow diagrams, architecture diagrams, and operational block diagrams.
- **Overleaf and LaTeX:** Overleaf and LaTeX will be used to prepare, format, and maintain the project proposal report and final documentation in a structured academic format.

CHAPTER 5: PROJECT SCHEDULING

5.1. Gantt Chart / Project Timeline

The development of the Karmasheel platform will follow an incremental approach from May 2026 to February 2027. After topic selection, proposal preparation, and literature review, the implementation will proceed in seven increments, beginning with essential modules and gradually adding advanced features.

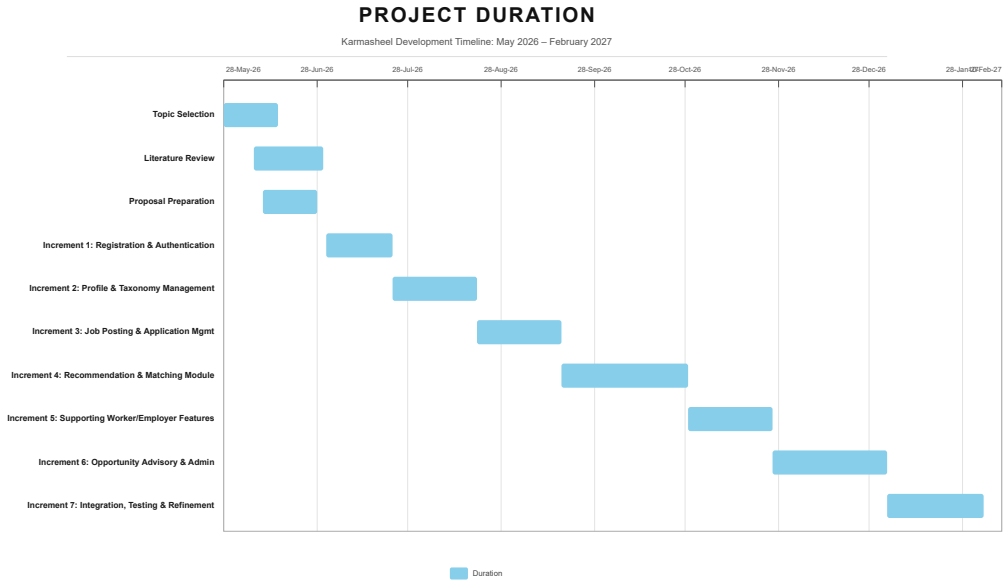


Figure 5.1: Proposed development timeline of the Karmasheel platform

CHAPTER 6: EXPECTED OUTCOMES

The proposed Karmasheel platform is expected to produce the following major outcomes in accordance with the project objectives:

- A functional web-based platform that supports worker and employer profiles, job posting, job searching, applications, employer verification, automatic CV generation, ratings, complaints, and admin monitoring.
- An explainable workforce-matching and opportunity-advisory system that provides worker-to-job and job-to-worker recommendations using NLP-assisted skill normalization, standardized skill tags, filtering, skill matching, location scoring, and practical opportunity-advisory suggestions.

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